

Kaway-Kawayan



Bamboo
Music
Education
Lessons
for K-12

Volume 1

High School

Kaway-Kawayan
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Volume 1
High School Set

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INTRODUCTION

“Sa Filipinas, ang kawayan ay malaking bahagi ng kultura dahil kaugnay ito ng iba’t ibang tradisyon, pagdiriwang, at paniniwala. Halimbawa, sa panitikan, ayon sa alamat, ang unang lalaki at unang babae ay nagmula sa isang pirasong kawayan. Sa sayaw, ang bantog na tinikling, singkil at subli ay isinasayaw gamit ang kawayan. Sa musika, ginagamit din ang kawayan sa paggawa ng iba’t ibang klase ng instrumentong pangmusika. Maging ang mga katutubong laro gaya ng luksong kawayan at palosebo ay gumagamit ng damong tropikong ito.”

(In the Philippines, bamboo plays a big part in culture because of its varied roles in tradition, celebration, and folk beliefs. According to Philippine folk literature, the first man and woman came from a bamboo pole split into two. Philippine dances such as the tinikling, singkil, and subli use bamboo accompaniment. Bamboo is also used in various types of Philippine musical instruments. Philippine games like luksong kawayan and palosebo use the tropical grass.)

(Source: <https://philippineculturaleducation.com.ph/kawayan/>)

In the distant past, our ancestors told a story about how the first Filipinos were born from the bamboo. The story goes that a bird was exploring an island when it heard a voice coming from a bamboo pole. Curious, the bird tapped on the bamboo. The bamboo cracked open and there emerged from it a man and a woman. The bird carried the two on its back toward an island where life as we know it began. At present, bamboo continues to play an important part in our lives. We use it as ingredient and tools in our cuisine, as medium in our arts and music, as building materials in our furniture and architecture, and as part of our decorative landscaping and flood and erosion control systems.

Like the first man and woman in the folktale, **Kaway-Kawayan** is a child of the bamboo, born from the labors of the Department of Science and Technology Forest Products Research and Development Institute (DOST-FPRDI), the UP Center for Ethnomusicology (UPCE), the Philippine Normal University (PNU), and the Philippine Society for Music Education (PSME). Together, these institutions and organizations worked on the Bamboo Music Instruments Innovation Research and Development Program, which is composed of projects seeking to develop the bamboo music industry in the Philippines through ethnographic and scientific documentation of folk and commercial bamboo makers, treatment of commonly damaged bamboo musical instruments, sound processing techniques and analysis, and bamboo processing and training center.

Kaway-Kawayan is a collection of music lessons on bamboo music integrating the content and performance standards of the Philippine K to 12 music curriculum. The lessons contained herein are a product of the collaboration of researchers from UPCE and writers from PNU and PSME. The ethnographic documentation of UPCE yielded significant records of bamboo music traditions in the Philippines, updating the research done by National Artist Jose Maceda in the 1970s until the 1980s. These evolving folk knowledge, skills, and practices documented in the ethnography of selected areas are imparted through the lessons written by the writers who were present during the fieldwork done by the ethnographers documenting the bamboo music traditions in situ or via online.

The first volume of **Kaway-Kawayan** focuses on the following Philippine bamboo music traditions and instrument makers: (1) Bamboo Organ of Las Piñas, (2) Bamboo Rondalla of Dipolog, (3) Bamboo Ukulele of Cebu, (4) Himig Kawayan of Quezon City, Laguna, Sta. Mesa; (5) Karatong Festival of Dulag, Leyte; (6) Musikong Bumbong of Angono, Rizal; (7) Musikong Bumbong of Bulacan; and (8) Tultugan Music of Iloilo. The lessons combine the ethnographic information contained in the bamboo music documentation with a sequential plan of musical experiences that are age-appropriate, culturally sensitive, and integrative of the Filipino children or youth's evolving musicking.

This volume has two sets: elementary (grades one through six) and junior high school (grades seven through ten). Content, standards, and competencies are derived from the DepEd K to 12 music curriculum. The music strategies reflect active listening, performing vocally and instrumentally alone or with others, evaluating one's music performance, and integrating bamboo music performances with related arts like speech choir, dance, or theatre. Materials used in the lessons were carefully selected from the ethnographic documentation and online video materials with accompanying links for media access. Reflections after each lesson enable the teacher to look deeper into the various lesson components and remediation activities when needed. Enrichment activities are also provided for learners who want to know more about bamboo traditions in general and bamboo instrument makers from the research sites in particular. When applicable, interdisciplinary activities for mathematics, social studies, physical and social sciences are included in some of the lessons.

Finally, the word "**kaway-kawayan**" is also a wordplay that means "bamboo waving." This wordplay is an apt metaphor for the swaying bamboo as it bends against adversity, never breaking but always adapting. We hope that through the lessons contained in **Kaway-Kawayan**, teachers, students, and bamboo enthusiasts can derive, interpret, and adapt educative experiences to keep the Philippine music like the bamboo – with roots firmly attached to our traditions and culture, but at the same time continuously growing and shooting upwards to face the needs and challenges of the present and the future.

Jocelyn Timbol-Guadalupe

BAMBOO ORGAN OF LAS PIÑAS

Prepared by: ANNA PATRICIA RODRIGUEZ-CARRANZA



I. OBJECTIVES

A. Content Standards

Demonstrate understanding of characteristic features of the Renaissance and Baroque period music

B. Performance Standards

Perform selected songs and dances from Renaissance and Baroque periods (dance suites: *canarios*, *allemande*)

C. Learning Competencies/Objectives

1. Listen perceptively to selected instrumental music of Renaissance and Baroque music
2. Explain the performance practice (setting, composition, role of composers/performers, and audience) during Renaissance and Baroque periods in the West and in the Philippines
3. Relate Renaissance and Baroque music to its historical and cultural background in the Philippine context through dramatization
4. Perform Renaissance and Baroque dances (can be accompanied by a chosen audio or on by playing on an available instrument) with correct pitch, rhythm, expression, and style
5. Describe musical elements of given Renaissance and Baroque music
6. Explore other arts and media that portray Renaissance and Baroque elements in the Philippine setting

II. CONTENT

Las Piñas Bamboo Organ: Its History and Music

III. LEARNING RESOURCES

A. References

1. Print Materials

BMI ethnography of Las Piñas Bamboo Organ

2. Audio-Video Materials

- a. BMI Video/Audio of the Bamboo Organ demonstration by Armando Salarza:

<http://tiny.cc/BMI-Bamboo-Organ>

- b. Video about Leo Renier

- c. BMI Pictures of the Bamboo Organ, its pipes and stops (as installed and in the museum)

3. Music Scores

Notation of the themes, melody, and bass lines of Salarza's demonstrations

B. Other Learning Materials

Classroom instruments (preferably piano/keyboard; but if none, any instrument capable of melody and harmony that the students already know how to play)

IV. PROCEDURE

A. Reviewing the previous lesson or presenting the new lesson

Show the students several photos of Catholic churches in the Philippines. Facilitate a discussion on how these churches were established in the Philippines, leading the class to talk about their previous lesson on the Spanish colonial period in the Philippines. Afterward, emphasize the picture of the St. Joseph Parish in Las Pinas and the Las Pinas Bamboo Organ. Ask if any of them is familiar with the history of the Bamboo Organ.

B. Establishing the purpose of the lesson

Tell the students that aside from being used for religious purposes, the bamboo organ is also used to perform for festivals, most of the repertoire is dance music from the Renaissance and Baroque period. Tell them they will perform the said dances.

C. Presenting examples/instances of the new lesson

Watch a video of Salarza's demonstrations on the pipe organ. Instruct the students to pay attention to the music and explanation regarding the canarios and allemande. Watch out for the meter and recurring rhythm of the dances. Afterward, create movements that are appropriate both to the musical and extramusical elements of the two dances.

D. Discussing new concepts and practicing new skills # 1

Group Discussion on Related Arts: Facilitate a discussion on Renaissance and Baroque music. Ask the students to do an elemental analysis of the two dances (and other examples that they may know of). Afterward, relate Renaissance/Baroque architecture, as seen in the photos of the churches presented at the beginning of the lesson. Guide the students in looking for parallelisms (e.g., ornate designs in the church, ornaments in the music as played on the bamboo organ) and help them deduce the socio-cultural context in which this music and architecture existed.

E. Discussing new concepts and practicing new skills # 2

Performance: guide the students in finding other recordings of *canarios* and/or *allemande* (can also be live music, if they know how to play instruments); brainstorming on a two to three-minute production skit incorporating dance reflecting musical sensibilities of Renaissance/ Baroque period in terms of movement, sound/music, costumes

F. Developing mastery (leads to formative assessment)

Create supervised practice and feedback: Have the students practice their presentation. Record each full run and watch out for particular difficult spots. Using distributed practice, identify difficult parts and practice in isolation.

Presentation: Have each group present their two- to three-minute original short skit, with a portion on dance suites (allemande and canarios), using their own dance, music, and instrumentation of found objects/classroom instruments, costumes, and props. Discuss how each performance was able to meet the objective.

G. Finding practical applications of concept and skills in daily living

Prompt: Art reflects the sensibilities of a particular period. How does today's music/art reflect society?

H. Making generalization and abstraction about the lesson

Prompt: Art can also fuel change in society. As a high school student, how would you use art to either reflect or change what is happening in society?

I. Additional activities for application or remediation

Differentiated Instruction: Students who have a hard time in performance may be involved in other tasks such as making costumes and props.

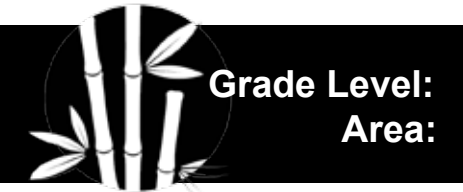
V. TEACHER'S REMARKS

VI. REFLECTION

- A. How many learners earned 80% in the evaluation? How many learners require additional activities for remediation?
- B. Did the remedial lesson work? How many learners have caught up with the lesson? How many learners continue to require remediation?
- C. Which of my teaching strategies worked well? Why did this work?
- D. What difficulties can my principal or supervisor help me solve?
- E. What innovation or localized materials did I use/discover which I wish to share with other teachers?

DIPOLOG BAMBOO RONDALLA

Prepared by: JOEY L. AYALA



I. OBJECTIVES

A. Content Standards

Demonstrate an appreciation and general understanding of rondalla music:

- history, origin of the instruments, the ensemble
- different instruments and their roles

B. Performance Standards

1. Follow specific instrument lines in a rondalla piece and sing the lines aided with sheet music (if available).
2. Perform in groups rondalla pieces (with or without instruments).
3. Arrange and perform in groups rondalla-style renditions of songs of their own choice or invention.
4. Demonstrate sensitivity to demands of ensemble performance (volume swells, dynamics, etc.).
5. Analyze complex issues (even non-musical ones) with the help of an analytical framework.

C. Learning Competencies/Objectives

1. Physical/Sensory: Make physical (as much as possible) contact with rondalla music and instruments. Experience playing or, at least, imagining playing). Appreciate the power of technologies such as music theory, instrument construction, arranging and conducting, etc.
2. Historical/Psychological: Have an idea of how rondalla became part of the Philippine musical ecosystem. Be aware of one's gut response to the ensemble's music and instruments.
3. Mythical/Symbolic: Have a sense of the potential power of the ensemble as a cultural element. Have a sense of how traditions can change through innovation.
4. Integrative/Unitive: Experience the flow state of ensemble playing. Experience being part of a larger body.

II. CONTENT

- A. Rondalla: Tradition, Innovation, Emotion
- B. An integrative experience and exploration of the rondalla ensemble

III. LEARNING RESOURCES

A. References

1. Print Materials
 - a. BMI ethnography of Dipolog's Bamboo Rondalla
 - b. Schematics of both bamboo and non-bamboo rondalla instruments (for comparison)
2. Audio-Video Materials
 - a. Videos of Dipolog's Bamboo Rondalla from BMI documentation:
http://tiny.cc/BMI-Dipolog_Rondalla
 - b. Videos of Jay Sarita's workshop from BMI documentation:
<http://tiny.cc/BMI-Sarita-Workshop>
3. Music Scores
Sheet music of rondalla arrangements featured in the videos

B. Other Learning Materials

1. Actual rondalla instruments, pictures, schematic diagrams, cross-sections.
2. Illustration of integrative analytical framework (suggested: Ken Wilber's AQAL - All Quadrants, All Levels. This is not necessarily for student use, but more for the teacher. This framework, among others, can be applied to any level and any subject.)

IV. PROCEDURE

Progress from objective contact with the material towards some form of understanding and application of the material. Given focused drilling, mastery may emerge. Technical mastery of the rondalla in a school year quarter is NOT the point of this module.

A. Reviewing the previous lesson or presenting the new lesson

1. Begin with experiencing rondalla music. Allow students to close their eyes and/or move-dance and/or draw – whatever the music makes them feel like doing.
2. Teacher draws out reactions and synthesizes the experience. (Suggested framework is Edward de Bono's Six Hats style of idea and discussion management. Cluster ideas into these 6 "hats": Facts or verifiable quantities, Feelings or strong subjective reactions, Optimism or the good things that may arise, Pessimism or the bad things that may arise, Alternatives or other options, Executive how to decide and what action to take.

B. Establishing the purpose of the lesson

1. The big takeaway here is to have a realistic view of culture and how it evolves using the rondalla as a case study.
2. The Rondalla Ensemble is an icon of Philippine music. Although its distinctive timbre is hardly heard in contemporary sound recordings, the tradition is alive and well in the context of schools, celebrations and public events that require some nod to the concept of Filipiniana.
3. What is its value to you as a listener? As a musician? Is it worthy of a significant investment of your time and effort? Is there joy in it? *Makakain ba yan?! What is its value on the global music stage? Why is it not on the MTV Channel? Should it be?*
4. Ideally the module may deepen to life-skill discussion such as:
 - a. How culture and technology are intertwined (For a diagram, see “Siningbayan: The Art of Nation-Building:” www.blafi.org).
 - b. Tradition, obsolescence, and survival of the most competent
 - c. Creating the future you want to see – like Jay Sarita, the bamboo rondalla instrument-maker.

C. Presenting examples/instances of the new lesson

1. Compare the sounds of traditional wooden rondalla instruments to those made of bamboo.
2. Try miming one short piece of each and feel the differences and similarities.
3. The teacher may use Six Hats to manage discussions - drilling students in this and other analytical frameworks sharpens and speeds up analytical processes. This is an asset across all fields of human endeavor!

D. Discussing new concepts and practicing new skills # 1

1. Imagining an activity stimulates the same neural and muscular pathways as the actual activity similar to visualization exercises done by top coaches and athletes.
2. Subjective oriented approach: Learners can practice a rondalla piece even without instruments by miming a piece - playing along and acting out their part in the ensemble. This may be funny at first and that's good and alright AS LONG AS the chosen piece is rehearsed (repeated over and over again until it feels presentable).
3. NOTE: Such an exercise (you can make up others) will give the experience of drilling-to-mastery even without the actual instruments and time to “burn in” the micro-muscle memory. These subjectively oriented exercises may be applied to non-musical life situations such as public speaking and going on dates. Imagining the situations handled with confidence will lead to actual confidence in actual situations.

E. Discussing new concepts and practicing new skills # 2

1. Objective oriented approach: Using videos of Jay Sarita's workshop, have the students inspect the physicality of the instruments. Look at the diagrams. Discuss the science and math involved in instruments and music.
2. Draw out and cluster ideas, deepen discussion to how Jay may be reshaping a tradition and how this relates to young people who are still deciding what to do with their lives. Is it safer to follow what others have done before you? What are the risks of innovation? (Advanced but probably inevitable discussion: Will technology make human beings obsolete?)

F. Developing mastery (leads to formative assessment)

Assumption if instruments are available: General guideline for all Mastery activities such as memorizing and rehearsing: The saying practice makes perfect is only true if you practice perfectly. This usually involves practicing slow and perfect progressing towards fast and perfect.

G. Finding practical applications of concept and skills in daily living

Prompt 1: Remember an inharmonious or problematic situation - perhaps an argument or confrontation. If the situation were a piece of music, how would you play your part in order to make the music sound better?

Prompt 2: When you are talking with someone be aware of your reactions to the other. Are you playing harmony or counterpoint? Are you speeding up or slowing down?

H. Making generalization and abstraction about the lesson

Sort out information, observations, feelings, criticism, etc., using this integrative quadrant by Ken Wilber, AQAL. (For a diagram, see "Siningbayan: The Art of Nation-Building:" www.blafi.org):

- Upper Right quadrant (UR): The **Individual Objective** quadrant. It holds details of objects - instruments, plectrums, strings, construction, materials, tools, costs, etc.
- Lower Right quadrant (LR): The **Collective Objective** quadrant. It holds details related to systems - the rondalla ensemble, the sum total of parts that add up to a complete instrument, the training of arrangers and players, the historical events that made the rondalla a Filipino orchestra, etc.
- Upper Left quadrant (UL): The **Individual Subjective** quadrant. Your feelings about rondalla instruments and music, any ideas, suggestions, desires and criticisms. Personal reactions or points of inspiration sparked by the materials, etc. May include critical appreciation of rondalla performances, arrangements, and musicians' skills and techniques.
- Lower Left quadrant (LL): The **Collective Subjective** quadrant. What role does rondalla play in Philippine arts and culture? Do people in general identify with the ensemble? How can rondalla strengthen your nation?

I. Additional activities for application or remediation

Work with real instruments for exploration, discovery, and eventual mastery.

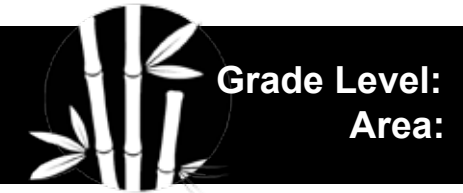
V. TEACHER'S REMARKS

VI. REFLECTION

- A. How many learners earned 80% in the evaluation? How many learners require additional activities for remediation?
- B. Did the remedial lesson work? How many learners have caught up with the lesson? How many learners continue to require remediation?
- C. Which of my teaching strategies worked well? Why did this work?
- D. What difficulties can my principal or supervisor help me solve?
- E. What innovation or localized materials did I use/discover which I wish to share with other teachers?

BAMBOO UKULELE OF CEBU

Prepared by: DOLORES TEXON-ANDRES



Grade Level:
Area:

10
Music

I. OBJECTIVES

A. Content Standards

Demonstrate understanding of characteristic features of Afro-Latin American music and Popular music

B. Performance Standards

Perform vocal and dance forms of Afro-Latin American music and selections of Popular music

C. Learning Competencies/Objectives

1. Listen perceptively to Afro-Latin American and popular music
2. Sing selections of Afro-Latin American and popular music in appropriate pitch, rhythm, style, and expression
3. Improvise simple vocal/instrumental accompaniments to selected songs
4. Identify the parts of a ukulele

II. CONTENT

Popular music and Cebu Bamboo Ukulele

III. LEARNING RESOURCES

A. References

1. Print Materials
 - a. BMI ethnography of Cebu Bamboo Ukulele
 - b. Photos of Cebu's Huni Ukuleles
 - c. Website of Cebu's Huni Ukuleles:
<https://huniukes.weebly.com/>
2. Audio-Video Materials
 - a. Huni Soprano Ukulele
<http://tiny.cc/YT-HuniUkuleles-Soprano>
 - b. Huni Tenor Ukulele
<http://tiny.cc/YT-HuniUkuleles-Tenor>

B. Other Learning Materials

1. Ukulele instruments
2. Basic Ukulele chord chart

IV. PROCEDURE

A. Reviewing the previous lesson or presenting the new lesson

Show pictures of trees: Rosewood, Langka, and Bamboo. Have a discussion on the products of these trees such as furniture, décor, and instrument.

B. Establishing the purpose of the lesson

Show pictures of bamboo ukuleles of Cebu's Huni Ukuleles. Explain that these parts can be made of different types of wood such as bamboo, rosewood or langka tree. Back panel of sound board can be made of bamboo. Fret board can be made of Rosewood. Langka tree can also be used as sound board.

C. Presenting examples/instances of the new lesson

1. Listen to Ukulele version of Ben & Ben "Leaves," Ed Sheeran's "Perfect," and John Legend's "All of Me." Do you know these songs? How popular are these songs?
2. Let the students explain what they like most in popular music. Is it the melody, lyrics, rhythm or emotional appeal?

D. Discussing new concepts and practicing new skills # 1

1. Song #1 Show the chord chart of "Perfect" C, G, D, em Practice the chord progression of the song. Use D D D DU D (D-down U-Up) strum pattern. Sing and play the song. Song#2 Show the chord chart of "All of Me" C, G, D, em, am
2. Practice the chord progression of the song. Use down strum pattern in syncopated rhythm.
3. Play and sing the song.

E. Discussing new concepts and practicing new skills # 2

1. Song #3 Show the chord chart of "Leaves"
2. Practice the chord progression of the song "Leaves".
Verse C-G-F 4x
Pre-chorus Am-D-G-C-F
Chorus C-G-F F-C-G Dm-Em-F G-C
3. Play and sing the given song.

F. Developing mastery (leads to formative assessment)

Presentation of the three songs by group.

G. Finding practical applications of concept and skills in daily living

Music is considered food for the soul. Make a playlist of songs for your family or friends so that they can be entertained and uplifted through music.

H. Making generalization and abstraction about the lesson

1. Young people like popular songs because of many reasons. It is easy to sing, easy to play, it has a catchy melody, and it is easy to memorize.
2. Let the students describe their performance in playing popular music and Ukulele.

I. Additional activities for application or remediation

Watch a medley of Cebuano folk songs accompanied by the bamboo ukulele. Sing and play some of the familiar songs together with the video.

V. TEACHER'S REMARKS

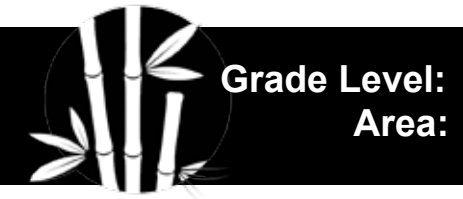
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VI. REFLECTION

- A. How many learners earned 80% in the evaluation? How many learners require additional activities for remediation?
- B. Did the remedial lesson work? How many learners have caught up with the lesson? How many learners continue to require remediation?
- C. Which of my teaching strategies worked well? Why did this work?
- D. What difficulties can my principal or supervisor help me solve?
- E. What innovation or localized materials did I use/discover which I wish to share with other teachers?

HIMIG KAWAYAN OF QUEZON CITY, LAGUNA, STA. MESA

Prepared by: PACITA M. NARZO



Grade Level: 7
Area: Music

I. OBJECTIVES

A. Content Standards

Demonstrate understanding of the musical characteristics of representative music from the lowlands of Luzon

B. Performance Standards

Perform music of the lowlands with appropriate pitch, rhythm, expression and style

C. Learning Competencies/Objectives

1. Identify the musical characteristics of representative music selections from the lowlands of Luzon after listening
2. Analyze the musical elements of some lowland instrumental music selections
3. Perform instruments/improvised instruments from Luzon lowlands

II. CONTENT

Instrumental Music of Pangkat Himig Kawayan

1. Angklung
2. Marimba
3. Bumbong

II. LEARNING RESOURCES

A. References

1. Print Materials

BMI ethnography on angklung, marimba and bumbong:

- a. BMI ethnography of Quezon City's Pangkat Kawayan
- b. BMI ethnography of Laguna's Kawayan Music Ensemble
- c. BMI ethnography of Sta. Mesa's Banda Kawayan

2. Audio-Video Materials

- a. Angklung Videos
 - Angklung – Himig Kawayan tutorial
 - Indonesian Angklung
 - Angklung Performance
- b. Marimba Videos
 - Marimba – Himig Kawayan tutorial
 - Philippine Music: Dipolog Marimba Nais Ko
 - Banda Kawayan Plays Pinoy Music
- c. Bumbong Videos
 - Himig Kawayan tutorial
- d. Pangkat Kawayan Videos
- e. Musikawayan /D' Original Musikong Bumbong 1896 Videos
- f. PUP Banda Kawayan/Banda Kawayan Pilipinas Videos
- g. Bocaue Central School Musikawayan Videos
- h. Musikawayan ng DepEd, Bocaue Videos

3. Music Scores

- a. “Rikiting-kiting,” arranged by Genaro Cesar Edilo Jr.
- b. “Pambansang Awit,” arranged by Genaro Cesar Edilo Jr.

B. Other Learning Materials

1. Piano Chords Tutorial Videos
2. Basic Articulation Videos

IV. PROCEDURE

A. Reviewing the previous lesson or presenting the new lesson

1. Discuss *legato* and *staccato* articulation through singing and listening to musical excerpts.
2. Review lessons on chord triads.

B. Establishing the purpose of the lesson

Tell the students that through the discussion on the Himig Kawayan ensemble, they will be able to:

1. Identify visually and aurally the basic instruments of the ensemble.
2. Explain the functions of the instruments and describe basically how the instruments are played.

3. Play with some of the instruments of the ensemble.
4. Identify some of the founders/ manufacturers of the instruments and develop appreciation for the Himig Kawayan ensemble.

C. Presenting examples/instances of the new lesson

1. Tell the students that they will be watching a video of a Himig Kawayan group. Ask them to identify and write down the instruments that they will recognize in the video presentation.
2. Let the students watch a video on the performance of any of the following Himig Kawayan groups:
 - a. Pangkat Kawayan founded by Victor Toledo
 - b. Musikawayan founded by Gilbert Ramos
 - c. Banda Kawayan Pilipinas founded by Siegfredo Calabig

D. Discussing new concepts and practicing new skills # 1

1. Ask the students to study the list of instruments that they made and to classify them according to the general classification indicated on the board. Let the students explain the basis of the classification on the board. The students should be able to identify and classify the following:
 - a. Angklung – idiophone
 - b. Marimba – idiophone
 - c. Bumbong – aerophone
2. Ask students to explain how the instruments are played as well as the functions of the individual instruments in the context of the ensemble as presented in the videos. Discuss that there may be other bamboo instruments tuned diatonically or other percussive bamboo instruments included in the Himig Kawayan ensemble as seen in the videos.
3. Let the students watch tutorial videos on angklung, marimba and bumbong for more information.

E. Discussing new concepts and practicing new skills # 2

Option 1 - If there are available Himig Kawayan instruments

1. Identify the founders/manufacturers of the Himig Kawayan instruments and discuss their contribution to Philippine musical tradition.
2. Distribute the pieces of angklung to the students. Demonstrate the manner of playing the instruments and other playing techniques. Let the students play the C major scale in legato manner then in staccato manner.
3. Guide the students in learning and performing “Rikiting-kiting” through any of the following techniques:
 - a. Reading the notes of the piece (if students are note readers)
 - b. Memory (through an assignment, ask the students to memorize the piece before this lesson on Himig Kawayan)
 - c. Teacher points at the players one note at a time

4. If the students can perform the angklung part on their own, play or ask anyone in the class to play the marimba part (and bumbong) as indicated in the piece. (The bumbong among the Himig Kawayan instruments is the most difficult to play. However, it can easily be learned through correct formation of the lips. Refer to the video on bumbong tutorial.)
5. If the students can perform the Himig Kawayan parts on their own, play or ask a student to play the guitar part in the piece.

Option 2 - If there are no available Himig Kawayan instruments

1. Identify the founders/manufacturers of the Himig Kawayan instruments and discuss their contribution to Philippine musical tradition.
2. Let the students watch more videos on:
 - a. Individual Himig Kawayan instruments
 - b. Himig Kawayan ensemble presentations

F. Developing mastery (leads to formative assessment)

Option 1 - If there are available Himig Kawayan instruments

Have the students perform the first part or the whole piece of “Rikiting-kiting” (depending on their skill or capacity). Let them practice playing the music, observing good phrasing and articulation with the end goal of achieving beautiful music (within the bounds of the time allotment for the class).

Option 2 - If there are no available Himig Kawayan instruments

Ask the students to identify the other instruments in the Himig Kawayan ensembles aside from the basic instruments discussed earlier. Let the students describe how the manner of playing and the material used in the instrument affect the quality of sound of the instruments.

G. Finding practical applications of concept and skills in daily living

Ask the students how playing together in an ensemble of angklung/ Himig Kawayan can be likened to the members of a family or to the different parts of our body in terms of function, contribution, or role.

H. Making generalization and abstraction about the lesson

Ask the students to: identify the basic instruments of the ensemble; describe the functions and the manner the instruments are played; identify the founders/manufacturers of the Himig Kawayan ensembles and their contribution to the Philippine musical tradition.

I. Additional activities for application or remediation

Option 1 - If there are available Himig Kawayan instruments

1. Assign students to their respective Himig Kawayan instruments and ask them either to finish working on “Rikiting-kiting”, part A and B (if the whole piece was not played during the class because of time constraints); or practice refining the whole piece with the correct interpretation.
2. Have the students perform the piece in the following meeting for a practical test (using rubrics on instrumental performance).
3. As a supplemental activity: Teach the students how to perform “Pambansang Awit” in another session.

Option 2 - If there are no available Himig Kawayan instruments

Assign the students to research on the history of any of the groups of Himig Kawayan listed below. For the next meeting, ask students to present their findings to the class.

1. Pangkat Kawayan founded by Victor Toledo
2. Musikawayan founded by Gilbert Ramos
3. Banda Kawayan Pilipinas founded by Siegfredo Calabig

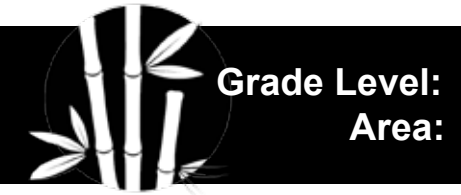
V. TEACHER’S REMARKS

VI. REFLECTION

- A. How many learners earned 80% in the evaluation? How many learners require additional activities for remediation?
- B. Did the remedial lesson work? How many learners have caught up with the lesson? How many learners continue to require remediation?
- C. Which of my teaching strategies worked well? Why did this work?
- D. What difficulties can my principal or supervisor help me solve?
- E. What innovation or localized materials did I use/discover which I wish to share with other teachers?

KARATONG FESTIVAL OF DULAG, LEYTE

Prepared by: JOCELYN TIMBOL-GUADALUPE



Grade Level:
Area:

7
Music

I. OBJECTIVES

A. Content Standards

1. Demonstrate understanding of the musical characteristics of representative music from the Visayas
2. Demonstrate understanding and application of musical skills related to selected Philippine festivals

B. Performance Standards

1. Perform selected vocal and instrumental music of the Visayas in appropriate style
2. Perform songs and dances from selected Philippine festivals

C. Learning Competencies/Objectives

1. Describe how music contributes to the performance of the musical production
2. Explore ways of producing sounds on a variety of sources similar to instruments being studied
3. Improvise simple rhythmic/melodic accompaniments to selected music from the Visayas
4. Evaluate music and music performances applying knowledge of musical elements and style

II. CONTENT

Karatong (Bamboo Slit Drum) Instrumental Ensemble and Festival from the Visayas

II. LEARNING RESOURCES

A. References

1. Print Materials
 - a. BMI ethnography of Dulag's Karatong Festival
 - b. BMI documentation pictures of the instruments used in the karatong ensemble

2. Audio-Video Materials

BMI documentation videos of Dulag 75th Anniversary Celebrations of MacArthur Landing featuring:

- a. Labiranan Festival of the 2019 Karatong Festival by San Jose National High School, Dulag, Leyte
<http://tiny.cc/BMI-Dulag-Karatong>
- b. "Commitment to Peace" speech choir presentation by the Youth of Dulag
<http://tiny.cc/BMI-Dulag-Choir>

B. Other Learning Materials

1. Classroom percussion instruments
2. Other found objects (canisters, plastic containers for drums, small bottles with beans as shakers)

IV. PROCEDURE

A. Reviewing the previous lesson or presenting the new lesson

1. Show a picture of MacArthur Landing and ask the students what they remember from the picture.
2. Show a map of Leyte and point out the location of Hill 120 in Dulag.
3. Let the students watch a clip of the speech choir during the 75th MacArthur Landing Commemoration at Hill 120.
4. Show a video/picture of the recent Yolanda / Typhoon Haiyan destruction of Leyte. Tell the students that the people of Leyte, Dulagnons in particular, have used music and art to cope with both man-made and natural disasters.

B. Establishing the purpose of the lesson

1. Show a picture of the karatong bamboo slit drum.
2. Tell them that they will see through the lesson how the bamboo slit drum was a symbol of Dulagnons' way of coping through storytelling, playing, singing, and dancing.
3. Tell the students that they can also apply these ways of expressing in found objects.

C. Presenting examples/instances of the new lesson

1. Let the students watch the video of San Jose National High School's Labiranan Festival, winner of the 2019 Karatong Festival.
2. Before showing the video, ask the students to watch out for the story of the presentation, the types of bamboo instruments, the musical patterns, and the movement used during each segment of the story.

D. Discussing new concepts and practicing new skills # 1

1. Divide the class into groups of 15 per group.
2. Give the students ten minutes to discuss how the elements of production (story, movement, music, instrumentation) were integral to the success of the presentation.
3. Using climate change mitigation as a theme, have the students brainstorm a two-minute dance drama presentation integrating story, music, dance presented using found objects and classroom instruments.
4. Assign task leaders for each aspect of production which may include story convener, narrator, and actors; music or soundscape conductor/facilitator, and performers; movement choreographer, and dancers).

E. Discussing new concepts and practicing new skills # 2

Conduct a workshop on the following topics:

1. Mind mapping
2. Introductory script writing for the story
3. Found object improvisation for the music
4. Improvised movement for the dance drama

F. Developing mastery (leads to formative assessment)

1. Conduct a supervised practice and feedback sessions using the following steps:
 - a. Have the students practice their presentation.
 - b. Record each full run and watch out for particular difficult spots.
 - c. Using distributed practice, identify difficult parts and practice in isolation.
2. Have each group present their two-minute short presentation on climate change mitigation using their own story, dance, music and instrumentation of found objects/classroom instruments.
3. Discuss how each performance was able to meet the objective of integrating ideas in story, dance, and music in one integrated performance.

G. Finding practical applications of concept and skills in daily living

Prompt: How can we express ideas through sound in everyday life?

H. Making generalization and abstraction about the lesson

Prompt: Music and arts have always been used by societies to express their daily life, triumphs, and struggles. How can I use music and art to help me as a present-day Filipino high school student in my daily life struggles and triumphs?

I. Additional activities for application or remediation

Use differentiated Instruction: Students who have a hard time in performance may be involved in other tasks such as recording the rehearsals for feedback using cell phone recorders if available.

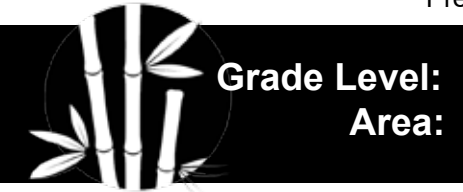
V. TEACHER'S REMARKS

VI. REFLECTION

- A. How many learners earned 80% in the evaluation? How many learners require additional activities for remediation?
- B. Did the remedial lesson work? How many learners have caught up with the lesson? How many learners continue to require remediation?
- C. Which of my teaching strategies worked well? Why did this work?
- D. What difficulties can my principal or supervisor help me solve?
- E. What innovation or localized materials did I use/discover which I wish to share with other teachers?

MUSIKONG BUMBONG OF ANGONO, RIZAL

Prepared by: DECEMBER RAGRARIO-VALENCIANO



Grade Level: 10
Area: Music

I. OBJECTIVES

A. Content Standards

Demonstrate understanding of the musical characteristics of representative music from the lowlands of Luzon

B. Performance Standards

Perform music of the lowlands (Rizal) with appropriate pitch, rhythm, expression and style

C. Learning Competencies/Objectives

1. Explain the distinguishing characteristics of representative Philippine music selections from Luzon in relation to its culture and geography
2. Sing a song from the lowlands of Luzon
3. Explore ways of producing sounds on a variety of sources that is similar to the instruments being studied

II. CONTENT

Musikong Bumbong of Angono

II. LEARNING RESOURCES

A. References

1. Print Materials
 - a. BMI ethnography of Angono's Musikong Bumbong
 - b. Pictures of Angono Musikong Bumbong from BMI documentation
 - c. Textbook pages
 - d. Optional: PowerPoint on History and Geography of Angono, Rizal
 - e. Pictures of Gigantes of Angono Fiesta
 - f. Clear map of Angono, Rizal

2. Audio-Video Materials

- a. BMI interview with Prof. Jose Rommel “Jopat” Gragera, conductor of Angono Wind Ensemble, and Angono Bamboo Instrument Maker
<http://tiny.cc/BMI-Gragera-Interview>
- b. BMI documentation of Marcelo H. Del Pilar students playing bumbong at Hiyas ng Bulacan 2020, Malolos, Bulacan
<http://tiny.cc/BMI-Bulacan-MHDP>
- c. Music videos from YouTube:
 - Video of Angono Wind Ensemble
 - Video of Magtanim Ay ‘Di Biro | Pinoy BK Channel
 - Video of Magtanim Ay ‘Di Biro by The Dawn
 - Video on How to Play the Rainstick
 - Video of DIY Rainstick (Fun Kid’s Craft)
 - Video of Italian making a rhythmic instrument out of bamboo

3. Music Scores

“Magtanim Ay ‘Di Biro”, a folk song of Rizal

B. Other Learning Materials

Available bamboo instruments, i.e., bamboo sticks, rainsticks, triangles, tambourines

IV. PROCEDURE

A. Reviewing the previous lesson or presenting the new lesson

1. Review on Dynamics as an expressive element in music.
2. Show a picture of Higantes.
3. Ask where they are popular and what they are associated with. (Higantes are popular during the fiesta in Angono, Rizal and are said to represent the artistry of Angono).

B. Establishing the purpose of the lesson

This is what we will learn about today:

1. About some information on the music tradition of Angono, Rizal
2. How the speed of a music or song is important in the way the music is performed
3. Singing and interpreting a song composed by a National Artist born in Angono

C. Presenting examples/instances of the new lesson

1. Today, by taking a look at a tradition practiced by our fellow Filipinos in Angono, Rizal, we will recognize values that we Filipinos live by: being community-based, musicianship, and spontaneity.

2. Sing a folksong from Rizal whose composer we do not know as it is a folk song.
3. In groups, create an improvised Rain Stick that we can use when performing the song, together with other available instruments.

D. Discussing new concepts and practicing new skills # 1

1. Discussion on the video on one of Angono's bands performing in a parade during the town fiesta:
 - a. In Angono, the fiesta is called Feast of St. Clement, the Patron of Fishermen, celebrated on November 23; The feast is said to have been made more festive and prominent by the wives of fishermen who helped their husbands sell fish to the people of the community. It is said that they would assemble in the patio or church plaza and join the procession to thank God for the blessings and bounty of the sea. These women were called Parehadoras. Their assembly for the procession inspired the attendance of musical bands, gigantes, and wet, muddy and dancing people.
 - b. The famous gigantes or giants composed of a Father, Mother, and Son represents Angono as the Arts Capital of the Philippines, as the saying goes the people of Angono learns to draw and paint before they learn to walk and talk.
 - c. Bands play a major role on this festive day where each school or barangay has a band that joins to help fill and keep the joyful mood and high spirit of occasion and for every fiesta occasion, at least 10 bands go marching and playing their music around the town.
 - d. The tradition of the famous bands of Angono is said to have started with a group of musicians who played musical instruments made of bamboo because they could not afford to buy those expensive brass instruments. These brass instruments were called Musikong Bumbong or Band instruments made of bamboo. With their strong musicianship, they created their own instruments out of the bamboo that was after all many in Angono. One of its makers was Lieutenant Lauriano Gragera. His instruments no longer exist, but there are a few displayed on the wall of a relative's house and are covered with cling-wrap for protection and posterity. These instruments are now said to be about 60 years old.
2. Using the photos and videos, compare the musikong bumbong instruments from Angono and Bulacan. Discussion:
 - a. The instruments seen in the photos and videos such as the bamboo baritone, French horn, and trombone are called Musikong Bumbong instruments which are famous in Bulacan and are in fact still existing and performed.
 - b. Since we are talking about music from Angono, there are however no more Musikong Bumbong instruments that still exist save for very few that are displayed in the home of the maker's family.
 - c. The difference between the Musikong Bumbong of Bulacan and the Musikong Bumbong of Angono is that those from Angono do not have the woven rattan bells but instead have buzzer endpipes.
3. Present and discuss some excerpts from BMI ethnography of Angono's Musikong Bumbong

4. Creating an improvised rainstick:
 - a. Mention that a simple bamboo instrument that we can make is the popular rainstick;
 - b. Add that rainsticks are made of bamboo but that looking for a piece of bamboo is difficult, hence homemade materials are used instead for the project.
 - c. Watch the following videos:
 - How to Play the Rainstick
 - DIY Rainstick (Fun Kid's Craft)
 - d. Present materials needed to make the improvised rainstick:
 - empty thick cardboard roll beans, small nails
 - small hammer
 - packaging tape
 - crayons/ribbons for decoration
 - e. Make the improvised rainstick
 - f. Practice playing the improvised rainstick in that it produces sounds shown on the video

E. Discussing new concepts and practicing new skills # 2

1. Present song “Magtanim Ay ‘Di Biro” via the videos:
 - Magtanim Ay ‘Di Biro | Pinoy BK Channel
 - Magtanim Ay ‘Di Biro by The Dawn
2. Learn song via Teaching a Song Method:
 - a. Listen to whole folk song
 - b. Brief discussion on some characteristics of folk songs:
 - talks on social issues;
 - learned through oral tradition;
 - intent of composer is obviously not for profit that is why composer is usually not known
 - c. rhythmic reading/syllabication by phrase
 - d. Learning the melody by phrase
 - e. sing the whole song
 - f. practice song in groups employing expressive elements

F. Developing mastery (leads to formative assessment)

In preparation for the assessment of a Creative Performance of “Magtanim Ay ‘Di Biro”, students will:

1. As a class, watch the video:
 - Italian making a rhythmic instrument out of bamboo:
2. In groups, perform the song “Magtanim Ay ‘Di Biro” employing:
 - Proficient rhythm and melody
 - Expressive elements and movement
 - Explore on texture with use of created improvised rain sticks and other available instruments, i.e., bamboo sticks, triangles, tambourines, and some form of application from the video
3. Assessment with teacher-made rubrics

G. Finding practical applications of concept and skills in daily living

Prompt: Since we were able to make homemade rainsticks, what other improvised music instruments can we make out of unused household items to manifest respect for the environment?

H. Making generalization and abstraction about the lesson

Sustaining the Filipinos' tradition of the fiesta is of value not only because it makes us a happier people but also because we become more sensitive and caring for one another as a community.

Filipinos' sense of musicianship includes not only the discipline of learning the techniques of performing the instrument, but also the skill of creating instruments out of the rich environment. More so, Filipinos' sense of musicianship includes the idealism of sharing the talent in fiestas with the community more than just performing on the pedestal of the stage to be revered by a few.

I. Additional activities for application or remediation

Create more improvised music instruments out of unused household items for addition to the texture as well as enjoyment to songs performed.

V. TEACHER'S REMARKS

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VI. REFLECTION

- A. How many learners earned 80% in the evaluation? How many learners require additional activities for remediation?
- B. Did the remedial lesson work? How many learners have caught up with the lesson? How many learners continue to require remediation?
- C. Which of my teaching strategies worked well? Why did this work?
- D. What difficulties can my principal or supervisor help me solve?
- E. What innovation or localized materials did I use/discover which I wish to share with other teachers?

MUSIKONG BUMBONG OF BULACAN

Prepared by: DECEMBER RAGRARIO-VALENCIANO



Grade Level: 7
Area: Music

I. OBJECTIVES

A. Content Standards

Demonstrate understanding of the musical characteristics of representative music from the lowlands of Luzon

B. Performance Standards

Perform music of the lowlands (Bulacan) with appropriate pitch, rhythm, expression and style

C. Learning Competencies/Objectives

1. Explain the distinguishing characteristics of representative Philippine music selections from Luzon in relation to its culture and geography
2. Explore ways of producing sounds on a source that is similar to the instruments being studied
3. Perform improvised instrument (French Horn) from Luzon lowlands (Bulacan)
4. Sing a song from the lowlands of Luzon

II. CONTENT

Musikong Bumbong of Bulacan

II. LEARNING RESOURCES

A. References

1. Print Materials
 - a. BMI ethnography of Bulacan's Musikong Bumbong
 - b. Pictures of Bulacan Musikong Bumbong from BMI documentation
 - c. Textbook pages
 - d. Optional: PowerPoint on History and Geography of Malolos, Bulacan
 - e. Pictures of ensaymada, pastillas wrapped in Japanese paper, longganisa, and buntal hat and bags
 - f. Clear map of Bulacan

2. Audio-Video Materials

- a. BMI interview with Rodrigo Anastacio, Bulacan Bamboo Instrument Maker
<http://tiny.cc/BMI-Bulacan-Anastacio>
- b. BMI documentation of Marcelo H. Del Pilar students playing bumbong at Hiyas ng Bulacan 2020, Malolos, Bulacan
<http://tiny.cc/BMI-Bulacan-MHDP>
- c. Music videos from Smithsonian and YouTube:
 - Video of How Brass Instruments Make a Sound
 - Video of How to Make a Hosepipe Horn
 - Video of Hosepipe and Orchestra
 - Garden Hose Concerto
 - Gaano Ko Ikaw Kamahal sung by Celeste Legaspi

B. Other Learning Materials

1. To make French Horn: water hose pipe, mouthpiece, funnel, packaging tape
2. (Optional) Real ensaymada and pastillas wrapped in Japanese paper

IV. PROCEDURE

A. Reviewing the previous lesson or presenting the new lesson

1. Show pictures of longganiza and ensaymada and say that this is their favorite breakfast. Also, show real/photo of pastillas and buntal-made hats and bags. Say that the best of such products come from the province of Bulacan, and that the bags may be better than the popular expensive Western bags (“proud to be Filipino” concept).
2. State where Bulacan province is, especially Malolos City, as this is the point for discussion for the day.

B. Establishing the purpose of the lesson

1. Ask the students: Where is Bulacan?
2. Show map.
3. Brief Discussion on Geography and History of Bulacan.
4. Focus on Malolos.
5. Discuss Bulacan as home to Filipino Patriotism and Music

Patriotism

- Philippines -first country to establish a constitution in Asia (Malolos Constitution was ratified in Barasoain Church during the Malolos Convention on Jan. 23, 1899) where our bill of rights emanated long before Asian neighbors made their own
- Bravery of Malolos women for whom Jose Rizal wrote a poem “To the Young Women of Malolos”

Music

- Bulacan is home to many of the country's greatest artists for music: Nicanor Abelardo, Francisco Santiago, Francisco Buencamino, Col. Antonio Buenaventura (National Artist in Music, 1988), Ernani Cuenco (National Artist in Music, 2000) and Alfredo Buenaventura; Queen of Zarsuela, Atang dela Rama (National Artist in Theater and Music, 1987); internationally acclaimed pianist, Cecil Buencamino Licad;
 - Music events in Bulacan are: Hataw sa Sining, a dance and music competition; Dulansangan, a street dance with music and drama. Pasiklaban ng Banda usually entitled "Konsierto ng Musikerong Bulakenyo" where bands all over Bulacan play their instruments together.
6. Tell the students: Today we will realize that the importance of Patriotism and music among the Bulakenyos/nyas still prevails as we learn about the:
- a. Unique band instruments made by Malolos resilient musicians
 - b. A song composed by one of the National Artists for Music who was born in Malolos Bulacan
 - c. In groups, create an improvised French Horn
 - d. Perform the song employing the improvised French Horn together with other available instruments for thicker texture.

C. Presenting examples/instances of the new lesson

1. Show pictures of Bulacan Bamboo Music Instruments such as Baritone, Saxophone, French Horn
2. Watch video on:
 - a. BMI interview with Rodrigo Anastacio, Bulacan Bamboo Instrument Maker
<http://tiny.cc/BMI-Bulacan-Anastacio>
 - b. BMI documentation of Marcelo H. Del Pilar students playing bumbong at Hiyas ng Bulacan 2020, Malolos, Bulacan
<http://tiny.cc/BMI-Bulacan-MHDP>
3. Reinforce particularly on how the bamboo instruments produce sound as per Mr. Anastacio.
4. Reinforce discussion on sound production of brass instruments through the video of How Brass Instruments Make a Sound

D. Discussing new concepts and practicing new skills # 1

1. Creating an improvised French Horn by watching the following:
 - a. How to Make a Hosepipe Horn
 - b. Hosepipe and Orchestra
 - c. Garden Hose Concerto
2. Present materials needed to make the improvised French Horn: Water hose pipe, mouthpiece, funnel, Packaging tape
3. Create the improvised French Horn making sure it produces a sound.
4. If project is done in groups, employ Integrative-Cooperative Approach by assigning each student in each group to do a specific tasks:
 - a. Rolling the hose pipe and putting them together with tape
 - b. Putng the funnel
 - c. Playing the improvised instrument

E. Discussing new concepts and practicing new skills # 2

1. Present song “Gaano Ko Ikaw Kamahal” via video:
“Gaano Ko Ikaw Kamahal” sung by Celeste Legaspi
2. Learn song via Teaching a Song Method:
 - a. Listen to whole song
 - b. Brief discussion on context of song composed by National Artist Ernani Cuenco from Malolos Bulacan Inspiration for composition is composer’s wife
 - c. Rhythmic reading/syllabication by phrase
 - d. Learning the melody by phrase
 - e. Sing the whole song
 - f. Practice song in groups employing expressive elements

F. Developing mastery (leads to formative assessment)

1. In groups, Perform the song “Gaano Ko Ikaw Kamahal” employing:
 - a. Proficient rhythm and melody
 - b. Expressive elements and movement
 - c. Improvised harmony from created improvised French Horn
 - d. Available instruments performed by ‘musicians’ in the group
2. Assessment with teacher-made rubrics

G. Finding practical applications of concept and skills in daily living

Prompt: Since we were able to make a homemade French Horn, what other improvised music instruments can we make out of unused household items so we don’t have to add to the pile of garbage?

H. Making generalization and abstraction about the lesson

Our learning today that Bulacan is home to Filipino Patriotism and Music, motivated us to enjoy creating an improvised French Horn and perform it together with a Bulacan-born composer’s song, and this made us experience the values of responsibility and sensitivity. Having a sense of patriotism and musicianship develops citizens with good character. The Greek philosopher Plato always knew this.

I. Additional activities for application or remediation

Create more improvised music instruments out of unused household items for addition to the texture as well as enjoyment to songs performed.

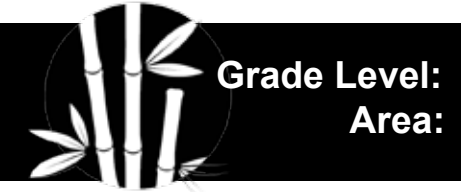
V. TEACHER'S REMARKS

VI. REFLECTION

- A. How many learners earned 80% in the evaluation? How many learners require additional activities for remediation?
- B. Did the remedial lesson work? How many learners have caught up with the lesson? How many learners continue to require remediation?
- C. Which of my teaching strategies worked well? Why did this work?
- D. What difficulties can my principal or supervisor help me solve?
- E. What innovation or localized materials did I use/discover which I wish to share with other teachers?

TULTUGAN MUSIC OF ILOILO

Prepared by: HERMIE F. CARTAGENA



7

Music

I. OBJECTIVES

A. Content Standards

Recognize and understands the salient features of the music of the representative music from the Visayas.

B. Performance Standards

1. Perform selected vocal and instrumental music of the Visayas (Iloilo) with appropriate rhythm, pitch, and style
2. Perform appropriate dance steps relevant to the Tultugan festival

C. Learning Competencies/Objectives

1. Describe the relevance of bamboo to the people of Maasin, Iloilo in terms of its culture and geographical set-up
2. Identify the characteristics of the Tultugan music
3. Explain the meaning of tultug and its socio-cultural function
4. Create and improvise melodic and rhythmic patterns based on Tultugan music

II. CONTENT

Tultugan Instrumental Ensemble and Festival from Maasin, Iloilo

II. LEARNING RESOURCES

A. References

1. Print Materials
 - a. BMI ethnography of Maasin's Tultugan Festival
 - b. Pictures of Tultugan instruments and performance
2. Audio-Video Materials

YouTube videos of performances in Tultugan Festival
3. Music Scores

"Cundansoy"

B. Other Learning Materials

1. Other Learning Materials
Improvised percussion instruments (empty bottles, cup, plastic containers, empty can, PVC pipes)
2. Remote Learning Resources
Improvised percussion instruments found at home (empty bottles, cup, plastic containers, empty can, casserole pot, PVC pipes, kitchen materials)

IV. PROCEDURE

A. Reviewing the previous lesson or presenting the new lesson

1. Show a picture of amakan and ask students about the process of making amakan and what material it is made of.
2. Show a picture of the municipality of Maasin and explain that the municipality is the bamboo capital in the Visayas.

B. Establishing the purpose of the lesson

1. Show a picture of tultug instrument and explain to the class its social and cultural function.
2. Show other Tultugan instruments like tala-tala, bayong, gabbang, paraspas, bamboo pipes, and mini bamboo pipes and elicit initial reactions from the students.

C. Presenting examples/instances of the new lesson

Watch a video of Tultugan festival and let the students observe the characteristics of its music in terms of melody, rhythm, texture, and style.

D. Discussing new concepts and practicing new skills # 1

1. Learn the song Cundansoy
2. Discuss the story in the song Cundansoy
3. Let the students explore various rhythmic patterns that may accompany the song Cundansoy.
4. Divide the class into three groups: group 1 will sing the song; group 2 will be the actors; group 3 will be the instrumentalists (improvised instruments).

E. Discussing new concepts and practicing new skills # 2

1. Let the students conceptualize a theme and make a story.
2. Using improvised instruments, students will create melodic and rhythmic patterns appropriate to the story.
3. Choreograph movements appropriate to the story.

F. Developing mastery (leads to formative assessment)

Supervised practice and feedback: Have the students practice their presentation. Give feedback to improve student's performance.

Group Presentation: Students present their output on Tultugan festival showcasing their original concept, dance movements, and musical ideas.

G. Finding practical applications of concept and skills in daily living

Prompt: How is sound reflected in our daily lives?

H. Making generalization and abstraction about the lesson

Music is a representation of one's culture.
The music reflects the way of life of a certain community.

Prompt: what's the impact of music on your life?

I. Additional activities for application or remediation

Let the students watch and observe Dinagyang performances. Let the students compare the Dinagyang and Tultugan festivals of Iloilo and identify their similarities and differences.

V. TEACHER'S REMARKS

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VI. REFLECTION

- How many learners earned 80% in the evaluation? How many learners require additional activities for remediation?
- Did the remedial lesson work? How many learners have caught up with the lesson? How many learners continue to require remediation?
- Which of my teaching strategies worked well? Why did this work?
- What difficulties can my principal or supervisor help me solve?
- What innovation or localized materials did I use/discover which I wish to share with other teachers?

Appendix A – Ethnographies

Excerpt from the BMI Ethnography of Las Piñas Bamboo Organ

“BMI Project: The Las Piñas Bamboo Organ”

By Juliet R. Bien

INTRODUCTION AND HISTORICAL BACKGROUND

To date, the Las Piñas Bamboo Organ is the oldest and the largest in the Philippines, and possibly the world (Tagle, interview, 9 February 2020). This musical instrument is the one and only functional organ made of bamboo and is actively used in liturgical services. What sets this apart from the other existing church organs is not only the material from which it is made but also its endurance because it has been around for more than two hundred years. The bamboo organ of Las Piñas is a cultural treasure, the only one of its kind in the Philippines.

At one point in the history of music, the organ was considered as the “king of instruments” because of its many musical characteristics together with its often grandiose presentation in churches. Its builder was not only a good musician, but also a universal genius with extraordinary talents as wood sculptor, metal-smith, painter, mathematician, and physicist. In the case of the Bamboo Organ, its builder was more than all that. Diego Cerra was also a natural scientist whose feat in curing the bamboo pipes was never successfully duplicated in decades and centuries to come. (Lauterwald, Helen Samson. The Bamboo Organ of Las Piñas, Manila: The Bamboo Organ Foundation, Inc., 102).

The Bamboo Organ was made by Father Diego Cera within the period of his assignment as parish priest of Las Piñas from 1795 to 1832. He began collecting the bamboo in 1816 and buried them in the seashore. The bamboos were unearthed in 1817 and by 1821, the organ was almost finished (Lauterwald, p.97). The long process of instrument-making required the cooperation of and assistance of local craftsmen who were already working with bamboo used for furniture and other household items (Tagle, interview, 2020). The exact date of installation is not known but it is believed to have taken place in 1824 when it was presumably completed.

The construction of the instrument was complicated because funding had to be sourced from Father Cera’s superiors and indispensable parts such as sheepskin, which were not locally available, had to be shipped. It was revolutionary and experimental at the same time because it would be made almost completely of bamboo. Historical records reveal that locals helped Father Cera in making the instrument.

In 1816, the bamboo was harvested and buried in the seashore. In 1817, Father Cera unearthed the bamboo and started constructing the instrument. Other wood types like narra, molave and kamagong were used for the mechanism as well. Simultaneously, the bellows were constructed, and sheepskin was shipped into the county. Last to be installed were the metal pipes which are believed to have come from Spain.

The entire set-up of the Bamboo Organ is like a house with several rooms installed in the wall of the Church. To access the instrument, one has to ascend the choir loft and descend a narrow flight of stairs. The organist plays from a small narrow

porch as if suspended but stable with the congregation approximately fifteen to twenty feet below. The chambers are on the opposite wall unseen to the audience but can be accessed by ascending a narrow wooden 5-foot ladder which can accommodate a single person. Upon reaching the topmost step, one enters a wooden room approximately 15 square meters which houses the mechanism, a hollow space surrounded by the old non-functional pipes and the intricate network of pipes from one end to the other. The keyboard levers could be seen on the first floor of the chamber together with the *pajarillo* mechanism, which has to be partially filled with water to provide the bird-like sound. Going up a few steps of a wooden ladder, one can see the other side of the bamboo pipes and the reed pipes.

The Bamboo Organ is now permanently installed on the upper left wall of the parish church of Saint Joseph, adjacent to the choir loft. As a celebration of heritage and sharing the instrument with the world, the Bamboo Organ Festival, a gathering of international and local musicians is held annually during the month of February. Repertoire for the organ is dedicated in these concerts which began in 1976 and continues to the present.

CEALWYN TAGLE: CONSERVING AND MAINTAINING THE BAMBOO ORGAN

Cealwyn Tagle's involvement with the bamboo organ began when he was a member of the Las Piñas Boys Choir when he was nine years old. Leo Renier, the choirmaster at the time, fostered his interest in the organ and encouraged him to take care of the instrument when Cealwyn was sixteen years old. Cealwyn had just graduated from high school when Mr. Renier referred him to Fr. Johann Trummer, a priest, organist, and a supporter of the Bamboo Organ Festival. Looking at the exquisite instrument as "orphaned" and exposed to neglect, Cealwyn accepted the job totally clueless of what the future had in store for him. Abandoning his desire to continue his college education, he bade his mother goodbye and left for Austria in 1988 to be an apprentice in organ making for four years. Fr. Trummer looked after Cealwyn, Armando Salarza (current Las Piñas Bamboo Organ resident organist), and other scholars from Las Piñas during their stay in Austria, and it was him who made their studies possible.

Cealwyn's interest in his new "job" fueled his desire to be in this field because of the country's lack of experts in organ conservators. His exposure to the European culture's utilization of organs in churches could have also increased his interest to promote the presence of organs in the numerous churches in the Philippines. After his four-year stint, Cealwyn stayed for two more years in Germany and worked for the company of Helmut Allgauer Orgelbau, where he was exposed to various fields in organ building. Here, his initial exposure as an apprentice was challenged with the intensive all-around work in the company. Cealwyn considers this period fortunate because, Orgelbau's company being small, he was always assigned to the difficult task of voicing the organs the company made. While improving his skills in woodworking, installations and repair, voicing enabled Cealwyn to hone his aural skills and craftsmanship. All these experiences have helped him in maintaining the Bamboo Organ and have also equipped him with the necessary skills needed in setting up an organ manufacturing company for himself upon his return to the Philippines.

Cealwyn came back to Las Piñas and was immediately assigned as the administrator of the Bamboo Organ. As administrator – a position he still holds to this day – he is tasked to conserve, repair, and maintain the organ. From the nine-year-old

chorister during the 1970's, Cealwyn is now in his 40's, a family man – married with three children – and the custodian of a 200-year-old natural cultural treasure, a post he never imagined occupying in his wildest dreams.

Cealwyn believes that Father Cera was sent to the Philippines primarily as an instrument-maker and secondarily as a parish priest. Father Cera most probably grew up as an organ builder before he was sent to the seminary and could have been sent to the Philippines to repair the organ at the Manila Cathedral. Aside from this, Father Cera also built the Baclayon Church in Bohol, whose flat façade feature is his signature style. It is also worth noting that Cealwyn is a direct descendant of Calixto Lara, one of the local citizens of Las Piñas who worked with Father Cera in maintaining the instrument together with his two brothers Pedro and Juan. The involvement of Calixto Lara's family with the Bamboo Organ now comes full circle as his great-great-grandson Cealwyn is now also in-charge of the Bamboo Organ after a 200-year interval.

Excerpt from the BMI Ethnography of Dipolog's Bamboo Rondalla

"Bamboo Music Instruments"

By Sol Maris T. Trinidad

USING BAMBOO

When Jay Sarita first ventured into bamboo xylophone-making, the reading materials he could find were only about making wood xylophones. Applying some of the techniques for wood, he started making his own bamboo xylophones. The first batch he made was an admitted failure. It was so bad that he wanted to just scrap the idea. However, he continued striving and found that the secret to using bamboo as a material was on the bamboo itself. He realized that as a material, it has very different characteristics from wood. To date, he still has not encountered a resource material for bamboo.

During the Tunogtugan Festival, Jay met Dr. Wu Shih-Yin and his Taiwan Bamboo Orchestra (TBO). Dr. Wu is a scientist who specializes in using bamboo as raw material for the instruments of the TBO. Jay learned many techniques on making bamboo instruments like the xylophone from Dr. Wu. When DCR went to perform in the Taiwan leg of the 5th International Rondalla Festival, Jay grabbed the opportunity to visit Dr. Wu's office. He spent three meetings with Dr. Wu in the latter's work laboratory, studying everything that he could in that short amount of time.

Jay also collaborated with the guitar maker, Adolfo Timuat. They discussed at length on the various characteristics of wood and acoustics. With Mr. Timuat, Jay learned all that he could about guitar-making. One of Jay's biggest takeaways was that no matter how meticulously one works with wood, the real outcome is not always what one reads; not all processes for wood are applicable to bamboo. He says that nothing beats actually working with the material and being open to asking questions and getting answers. The products that he produces are a result of his own experimentation.

When working with bamboo, Jay has to take the nodal points of the bamboo into consideration: the closer to the node, the denser or thicker the material. This makes its flexibility incomparable to wood, i.e., a 2-inch-by-4-inch-sized wood would have an even grain. With a bar of wood, the left and right ends are very similar, almost uniform. With bamboo, it is very different and looks like it is all over the place. Some bamboo is also crooked both laterally and longitudinally. It is difficult working with bamboo.

As of writing, *buntong* or *botong* (*Dendrocalamus latiflorus*) is the ideal material for Jay's instruments. Other varieties of bamboo available in Dipolog like the *lunas* (*Bambusa nguyenii* Ohrenb.) and *tungkan* (*Bambusa blumeana*) proved to be too thin for his needs. Buntong or botong can be found eight kilometers away from Dipolog City, both in upland and low-lying areas. The farther from the city, the more of this variety may be found, as long as the area is near a river. Jay has a regular contact person from whom he sources this bamboo. He entrusts the choice of quality bamboo to this person who is of advanced age. He also likes that he purchases them at a reasonable price.

The schedule for harvesting the bamboo is left to his contact as some of the considerations are the shifting of the tides and the like. When ordering bamboo, he has a waiting time of one week. But Jay does not use this batch of bamboo yet. The stock that he used recently in making his prototype were bought just after the 2013 Tunogtugan

Festival. He air-dried and stored these 12 pieces of 8 feet long poles, upright, on a concrete flooring in his backyard. He made sure to treat his stocks with termite spray, Solignum. He has used a lot of this bamboo batch from 2013.

Save for naturally air-drying the bamboo for long periods, he has yet to use other drying techniques like smoking. Thus far, he has only purchased bamboo twice: in 2013 and most recently in 2019. He avoids keeping too much stock as it might decay or be infested and become unusable.

Jay can make approximately 10 units or more of a 3-octave chromatic xylophone from 12 bamboo poles. While the bars are made from botong, the frame is made from other thinner bamboo varieties like tungkan and other local ones. When Jay would need a 1x2 cut size, he would do two layers using the botong but up to five layers of the other bamboo types.

Before, Jay used bamboo only for the bars. The resonators were usually made of polyvinyl chloride (PVC) pipes as PVC is easy to work with. Recently, however, Jay and his workers were able to process bamboo into square-shaped cylinders as resonators. At first, this was difficult because when they cut the bamboo with a bolo, it did not come out straight like wood; later, they discovered that the safest and straightest way to cut the bamboo into square resonators is to handsaw it.

To make bamboo resonators, they first draw a line on one side and a parallel one on the other side using a ruler and a pen. These lines serve as their guide for sawing the bamboo with a bow-type hand saw. To get the desired width, they process the cut bamboo using a thickness sander. With this machine, Jay can dictate how thin the bamboo will be, even up to veneer size. All the instruments made in their shop go through this sander. This sander was made by guitar makers. He bought it from them for Php15,000 plus shipping fee of approximately Php 3,000.

The decision to change the resonators to square from cylinder was made due to the difficulty in sourcing the appropriate bamboo dimensions to match the thickness of the xylophone bars. This also gave him the upper hand in presenting that they have the capacity to make resonators from both PVC and bamboo, cylindrical and square. He also saw that the bamboo xylophones available in the market only had the frames made of bamboo with the bars made from narra and the cylinders from PVC. So, when he says that they have bamboo xylophones, he can rightfully claim that everything about it is made from bamboo.

One of their products is the table-top xylophone. This is unique in that the resonator is the travelling case of the instrument. He designed it because it made the xylophone easier to carry. It can even be hand-carried into an airplane due to its compact body. He started selling this in 2017.

Prof. Jocelyn Guadalupe shared during the interview that Atty. Dulce Punsalan was able to show her the table-top xylophone prototype from Jay. The attorney found the instrument to be very durable between various climate conditions as she had brought it with her in international trips. The tuning did not also change. (Prof. Guadalupe said that the drawback with bamboo was that the tuning can change depending on the temperature in the area.) Jay attributed this durability to the botong bamboo material.

BAMBOO RONDALLA

The *rondalla* instruments that Jay has made from bamboo are comprised of the 5 *bandurria*, 2 *octavina*, 2 *laud*, 2 classical guitars, 1 bass, and 1 marimba. He differentiated between the regular *bandurria* and what he called the RS model (named after National Artist for Music Ramon P. Santos) of which he made one. The *octavinas* differ from the regular guitar-shaped *octavina* in that it is tear-dropped shaped. The design of the *laud* also had a different approach. It took Jay and his workers three months to make all 13 instruments.

The length of aging process for the bamboo used in this particular set of *rondalla* instruments is the same as that of the xylophone. The bamboo also came from the same clump of bamboos. However, the xylophone made use of a much older bamboo. As the bamboo they used was already available, the processing time did not include the bamboo preparation time.

Making of a bamboo *rondalla* was a standing challenge from Ms. Aurea Lopez after she had visited his workshop in 2014. She asked if it was possible to make a *bandurria* from bamboo. Because Jay answered that it was possible, she then asked him to make one. The idea, however, was shelved and forgotten for five years. Later, he got married and became less active in the *rondalla* performing scene. One day, he asked one of his workers about making a *bandurria* from bamboo. His worker answered that it was easy to make one and that they could start building it the next day. So, Jay, with four of his workers, started studying the mechanics of making the instrument.

One of the questions they had was on the ideal process of binding the bamboo. At that time, he had no idea about veneers. When thin enough, bamboo can be torn by hand. What Jay found out was that companies like Yamaha or those in China used veneer for their guitars but arranged them cross-grain. They made the observation that the sound of solid wood was louder than instruments made from the cross-grain veneer. This made Jay think of not using the cross-grain method in binding bamboo. There was no available information on working with bamboo, but the ready-made cross grain veneer instruments already produced weak sound boards.

Jay decided to make things differently by doing a same-grain two-layered veneer for their *rondalla* materials: The veneer was cut in approximately 1-inch widths and joined together as the 1st layer. The second layer was also made of the same veneer widths, but the joinery of the 1st layer was in between the joinery of the 2nd layer. In the event that one layer fails, there was still another layer that provided support. This resembled the idea of hollow blocks, alternating or interlocking. After making their first bamboo soundboard, they compared it with a spruce sound board. Bending both laterally produced the same results. Cross-grain veneers could not be bent in the same way. And so, they continued with making the instrument.

The first one they finished was a *bandurria*. However, Jay wanted to test the sound immediately, so even before the finished *bandurria* got varnished, Jay had already attached the strings. He observed that the lower registers sounded just like the *bandurrias* made from wood, but the higher register notes had a thin quality about them that was unsatisfactory. So, they took the instrument apart and further thinned-out the sound board. The new sound board produced an acceptable sound and they continued making the rest of the instruments.

In coming up with his bamboo instruments, Jay did not visit other manufacturers anymore. His collaborations were with other *rondalla* musicians like Ben Brillantes, a luthier from Dumaguete, and William Alama, who was also involved in the Gitara Ni Juan Project of the Department of Science and Technology. The same-grain veneers were entirely his idea that he and his workers put into work. For their instruments, they adopted the coconut shell as material for the finger boards from Mr. Alama. Any ba-o ng niyog can be used, no particular coconut variety was identified. The shell was cut into 1 cm x 1 cm or 15 mm x 15 mm tiles for their use. Jay found the results very beautiful in sound. While he had asked permission from Mr. Alama in also using coconut shells, the process that Jay's workers put it through is different from Mr. Alama's. During a visit at Jay's workshop, Mr. Alama commented that their processing time was shorter, and Jay's workers finished faster.

All the wooden materials used in the instruments of the bamboo *rondalla* are from bamboo except for the neck, which is made from mahogany. The sound board, side, and back are from bamboo. Some of their finger boards are made from the coconut shell while the others are from local wood like mahogany. The bracing of the *bandurria* is also from wood. Even if Jay wanted the finger boards to be made from bamboo, he was concerned that the fibrous quality of the materials will injure the musician. This made him decide to use wood or the coconut shell as finger boards.

For comparison, they used bamboo for the bracing of one of their classical guitars and wood for the other one. For comparison also, one classical guitar used wood for the front and back. Jay observed that the guitar using only bamboo all over was better sounding to his ears than the one that had wood mixed in. Jay surmised that the frequency of bamboo was united in the all-bamboo guitar.

With the laud he made, they changed the position of the f hole and computed the load of the bridge to make it hold the instrument despite the effects of changes in the weather on the instrument. Jay also decided to change the design of the octavina to make the *rondalla* family of instruments the same size and visually similar. Despite the new shape of the octavina, Jay states that it is still the same octavina instrument because the scale length is the same. This made Jay interested to adopt the tear-drop shaped bass that Mr. David Dino Guadalupe showed him.

For the players of the instrument, the bamboo instruments are heavier than the wooden ones, especially the bass. However, there have been no complaints from the players so far in terms of action, playability, or even how they hold the instrument or seat with it. Some did not even notice that they were already using the bamboo *rondalla*.

If compared, the sound of his bamboo and wood *rondalla* are close. At the moment, if their concert *bandurria* has a 10 rating, with 1 as the lowest, the bamboo *bandurria* prototype is at 8. In terms of cost of materials, working with bamboo is a whole lot cheaper than working with wood, especially spruce. However, in terms of manpower hours, bamboo instruments cost a lot more. A final costing has not yet been made as of writing.

At the moment, there are no noticeable storage or care requirements as these instruments were just recently finished. Jay is waiting for any of them to fail as this will help in improving the instrument.

Some makers just build the instruments without thinking. Jay's advocacy is to come up with ways to make the instruments more professional-looking with a beautiful sound – the whole package – for as long as the resources are available and accessible. Jay says that if there is something we can contribute to this, then we should. The ultimate purpose is to raise the standard of the *rondalla* instruments.

Excerpt from the BMI Ethnography of Cebu's Bamboo Ukulele

"The Bamboo Ukuleles of Cebu: A Trailblaze"

By Yves P. Cuenza

THE UKULELE BAMBOO INSTRUMENTS

There are different factors as to why a lot of people (both musicians and non-musicians) tend to get attached to playing the ukulele. Mr. Brian Brusas of the Huni group asserts that the advantages of the ukulele instrument include its being small and handy. Moreover, the laid-back perception that it gives does not intimidate or give pressure to the musical ability of the person playing it.

The *Bamboo Musical Instruments (BMI) Description Table* below illustrates the differences of each ukulele:

Instrument		Method of Sound Production	Size (scale length)	Number of frets	Tuning	Role in the Ensemble
ukulele	sopranino	Ukulele is a lute instrument that is plucked and strummed	28 cm	10	d1 g1 b1 e2	It can be used as a solo instrument, for accompaniment, and for ensemble music making.
	soprano		35 cm.	12-15	g1 c1 e1 a1	
	concert		38.5 cm	15-20	g1 c1 e1 a1	
	tenor		43 cm.	15-20	g1 c1 e1 a1	
	baritone		51.5 cm	18-19	d g b e 1	

The Huni group produces five types of ukuleles. The *soprano*, *concert*, and *tenor* are the most commonly used ukuleles which are tuned on the same key but gives give different over-all timbres. The difference is usually seen on the body size which is proportionate to the sound it gives off – the bigger the size, the deeper and fuller the ukulele would sound. The baritone is the largest among the ukuleles and it has the same tuning with the upper four strings of the guitar. The smallest is the *sopranino* and has the same tuning with the baritone but is set one octave higher. The price of every bamboo ukulele ranges from P5,000 to as much as P10,000, depending on personalized requests from the buyer

The shop has been producing handmade bamboo ukuleles. The most preferred species as raw material is locally known as *bontong*. If a customer wishes to get a bamboo ukulele, they must know the advantages and disadvantages of having one. Mr. Brusas and his team are quite upfront and honest in presenting those terms. Probably the only disadvantage about the sound is that the bamboo material does not necessarily produce the brightest timbre compared to the more traditional materials like mahogany, mango and *langka* woods. On the other hand, one of the strongest feedback that the bamboo material has been getting comes from foreigners, who say that the bamboo ukulele is more adaptable to climate and seasonal changes wherever the instrument is taken.

Excerpt from the BMI Ethnography of Quezon City's Pangkat Kawayan
Musiko Bumbong of Pangkat Kawayan: The Glory of the Singing Bamboos of the Philippines
By Janine Josephine Arianne A. Liao

A LIFETIME GUARANTEE: ELEVATING THE HUMBLE BAMBOO TUBE

Under the creative leadership of Victor Toledo, the old Pangkat Kawayan of Quezon City produced music using the bamboo instruments that were created, designed, and refined by the master himself. The current Pangkat Kawayan uses the same instruments that were made at least thirty years ago but have aged beautifully with the group. With the dedication and attention to detail evident in their quality, these instruments attest to the standard of a lifetime's guarantee that can only come from meticulous research and production

One important aspect that I have noticed is that a major driving force for the creation of these instruments was for the bamboo instrument to be able to play a western, diatonic scale. In an article, Vic Toledo writes that the *bumbong*, or tuned bamboo tube, needs to play a complete scale, in tune, in order to make a beautiful sound. According to him, most bamboo instruments in the Philippines are "not tuned," that is to say, pitches are indefinite in comparison to western tuning. This undertaking to tune the bamboo seems to stem from a very cosmopolitan outlook, indicating a desire to participate in a global music culture. It also indicates a strong local thrust in developing distinctively Filipino instruments through utilizing and perfecting bamboo instrument making and design.

Among the instruments that the Toledo makers crafted, the most unique and significant is the *bumbong*. These are thick in diameter and appear to be quite heavy. With a hole at the side, it appears as a thick transverse flute. However, unlike the thin flutes with multiple holes, the *bumbong* produces a single pitch per tube. They have a complete set of two and a half octaves. As Vic writes, it is a unique contribution of the Philippines to the world.

Another instrument that looks unique is the *tipanklung* or bamboo piano. It is similar to the *angklung*, or bamboo rattles from Indonesia. However, instead of these played separately by different individuals, the rattles are hanged on a frame and arranged like a piano, including the raised black keys. It looks like a curiosity, but once it is hanged like so, it is possible to be played by one person. When they performed for us, this instrument carried the main melody. This instrument is one of the later creations of Vic Toledo.

They also have an instrument that they call the Philippine *angklung*, the *alugtog*, which is played similarly by groups in an interlocking style. However, the appearance of the instrument features a different design, the most notable of which is the three-line staff with a black sixteenth note in the center. This appearance follows the western model and is a mark that signals its easy adaptability to western-style music.

Another form of bamboo rattles are the chord rattles. What is unique about this instrument is that each rattle is already made up of three pitches, forming a chord, which can be played by one person as accompaniment. The members who demonstrated these performed with choreography, with simple turns and forward-backward sways.

They also have a marimba, whose bars are made of *narra* and resonators made of bamboo instead of metal. The marimbas in the group play melodies, harmonies, and obbligato fills in the music arrangements. I noticed that the senior members of the Pangkat Kawayan were the ones who played the marimba.

An interesting instrument was lying at the corner of the Pangkat center. According to Beth, it was not used anymore. It was the *tunog hagdan* bamboo instrument. And true enough to its name, it looks like a ladder. The “steps” are made of narra bars, and can be classified as a xylophone instrument. Beth demonstrated the instrument for us by performing the song Pipit which was arranged by Vic Toledo. This instrument is unique to Pangkat Kawayan and is said to be one of Vic’s many musical instrument inventions.

Lying at the sides of the Pangkat center were the *gabbang*. The *gabbang* is a xylophone found among indigenous people in southern Philippines. However, according to Chino, these were not tuned and were therefore not used at all.

Finally, a curiosity of a piece was the sungkit. There was only one available, and as they said, it was for aesthetic purposes, and did not function for sound at all. It was a long piece of bamboo, similar to a sungkit or a device used to get fruit from trees.

The Toledo *bumbong* instruments have been around for a long time and are cleaned with alcohol as they have been shared all these generations. But these are indeed of good quality – well-aged, solid-sounding, and sturdy. For minor repairs, they just apply white glue, and for fine-tuning, a small piece of wood is attached to the inside, and I believe this small piece can then be manipulated to alter the tuning of the *bumbong*.

Currently, the sets that they use are stored in the Pangkat Kawayan center, but they can be easily packed for travel. Even the *tipanklung* can be disassembled for easy shipping. Aside from the ones used in performance, there are still some instruments around. These spares are stored in the Pangkat Kawayan center, but some are still at the Toledos’ house. There also may still be broken sets of angklung, still untuned. Sadly, those without storage space have been thrown away.

Meanwhile, a different form of invaluable wealth can be found in the Pangkat Kawayan Center: posted on the walls are articles on bamboo written by Victor Toledo himself. One of these articles is a collection of excerpts from a Japanese book on bamboo that he translated. Another is a treatise on the history of the *bumbong* (bamboo tube) as a cultural item and a musical instrument in the Philippines that he co-researched with a Vicente Toledo, professor of Spanish language and music in University of Manuel L. Quezon. Most noteworthy is this article that provides direct insight into the manufacturing process that Vic Toledo had mastered. An outcome of his personal experiences, research, observations, and consultations with builders and carpenters, this article discusses in detail the process of choosing and treating of bamboo, as well as cutting it and putting the pieces together to form instruments.

First, Vic shares that it is best to cut bamboo during summer. In the rainy season, the bamboos have new shoots, and these young sprouts have a “sweet” essence that attracts *bukbok* (rice weevils) which damage the bamboo. If cutting bamboo during the rainy season is unavoidable, it is best to soak it in vinegar in order to counter its “sweetness” and prevent it from being attacked by *bukbok*. Once the bamboo is dried,

wood treatments like Solignum can be used. Wood varnish – regular wood varnish, according to Vic’s son Chino – was also applied to the *bumbong* found in the Pangkat Kawayan Center, but Vic does not mention this in his articles. Also discussed is the use of sandpaper as a seal of sorts and how it provides better resonance, possibly due to the rough surface. This may be a unique practice to the Toledo makers.

Next, Vic warns against using old or mature bamboo. Older bamboo tubes, which can be identified by their telltale reddish tone, easily break and do not have the ideal sound. He says that the ideal age for bamboo for instrument making would be around five years, when bamboo possesses sturdy and strong fibers and a clear, bright sound. Clearly, he also cautions against using cracked, bent, or crooked bamboo tubes, as these cannot be tuned properly. He further shares that tropical bamboos have harder fibers, thus producing a bright and clear sound as opposed to bamboos that grow in more temperate regions, which in turn have softer fibers, therefore producing a softer and unclear tone.

Vic shares an interesting observation about how to tell if the bamboo is cut in the morning or at noon just by examining the “buko” or “eye” of the bamboo. He says that the *buko* is “closed” in the morning, but gradually opens up at noon time. He continues that that it is a trait of tropical bamboos but does not go any further to tell what this information may be used for. However, he enumerates four factors that affect the tuning of the *bumbong*: 1) the thickness or thinness of the bamboo; 2) the hardness and age of the bamboo; 3) the length, diameter, and the actual hole of the bamboo; and 4) the location or position of the hole for blowing into the tube. He emphasizes the importance of where the blow hole is bored into the bamboo tube as it affects the exactness of the pitch.

Chino recalls amusing stories of how his father, almost mesmerized by the beauty of the bamboos, would point at bamboos along the highway and remark if these were suitable for instrument making or not. This is truly a mark of the master craftsman who knows his materials well. In his articles, Vic mentions bamboo kalabaw (more commonly known and also referred to as kawayan tinik in the Pangkat Kawayan Facebook page). He refers to it with the scientific name of *bambusa blumeana* and says that is the best bamboo for making instruments. Vic recommends this species because it possesses the right thickness, ideal fiber density, and a bright sound, making it easy to tune properly. Initially, the Toledos sourced their *bambusa blumeana* from their hometown in Nueva Ecija, where right behind their old house bamboos grew abundantly, but for a time, they were also sourcing bamboos from Montalban.

Bamboo, however, is not the sole material for the Toledo instruments. *Narra* wood, for example, was used for their xylophone slabs, particularly the marimbas. Chino explains that slabs made from narra wood give a good sound and at the same time, are sturdy and do not break so easily. According to him, they also tried *yakal*, but it had no sound at all. Yet bamboo is still a necessary material for the marimbas, as it is used as resonators to amplify the sound of the slabs. A quick glance shows these bamboos to be of uneven length. The main reason for this is that the nodes, and not the length or entirety of the bamboo, are the ones tuned to the slabs. This is why, according to Chino, bamboo tubes are cut from node to node. In fact, when bamboo tubes don’t have a node – as seen in some of the bamboo resonators – their bottoms are sealed off with an artificial cover to simulate a node so that they can be tuned to the necessary pitch. For some of the lower pitches, tuning the tube is not as important, but the tube is still

necessary for increasing volume. As for the other resonators, some are there just for aesthetic purposes, in order to fill in the gaps in between the actual bamboo resonators.

When asked about what the main innovation contributed by his father and grandfather could be, Chino said that it is not in the instrument design, but rather in the process of making the instrument. Truly, the meticulous process is astounding in itself and is the main reason for their quality and durability. As his son Chino would say, Vic Toledo's efforts in perfecting bamboo instrument making and design stem from his passion in treating the bamboo not as a toy or souvenir but an instrument built to last a lifetime.

Excerpt from the BMI Ethnography of Laguna's Biñan Kawayan Music Ensemble
"Ret. Col. Gilbert Ramos, Biñan Kawayan Music Ensemble, and Musikawayan"
 By Roan May D.G. Opiso

BAMBOO MUSICAL INSTRUMENTS

Mr. Ramos and his family made the following bamboo instruments that are used by the different ensembles that he leads.

Instru- ment	Method of Sound Production	Melodic Range	Role in the Ensemble	Function in Ritual or Social/Cultural Activity
<i>Angklung</i>	Sound is produced by shaking the instrument	1 pitch per instrument	Plays melodic line; fill in the chords	An ensemble, the instruments are played during fiestas and other celebrations, mostly functioning as entertainment music.
<i>Piangklung</i>	Sound is produced by shaking the instrument	Around 2 octaves	Plays lead melodic line	
<i>Bumbong</i>	Sound is produced by blowing air into the hole of the instrument	1 pitch per instrument/ bamboo tube	Functions Like a horn and tuba in lower pitch	
<i>Rambong</i>	Opening of bamboo tube is struck using rubber mallets	One and a half octave	Functions like a bass	
<i>Marimba</i>	Wooden keys are struck using mallets	2 octaves	Plays lead melodic line and counter melody	
<i>gabbang</i>	Wooden blades are struck using mallets	2 octaves	Plays melodic line and effective for marching	
<i>Pito</i>	Sound is produced by blowing air into the hole of the instrument	2 octaves	Plays lead melodic line and counter melody	
<i>Bamban</i>	Sound is produced by humming (like a kazoo)	Conforms the range of the voice of the player	Plays lead melodic line	

Angklung is a bamboo shaker which is an instrument that is tuned to different pitches (Angklung 2013). One person plays one angklung which corresponds to one note.

Piangklung is a set of angklung hanging on a wooden frame. It is arranged like a piano keyboard (Piangklung 2013) and is played by one person. This instrument plays the melody.

Bumbong is a blown bamboo tube instrument positioned transversely from the player's mouth. In the Biñan Kawayan Music Ensemble (BKME), it is grouped chromatically in twos or threes and functions like a trombone and tuba in a marching band.

Rambong is a set of 13 bamboo tubes positioned vertically in a frame. The player hits the opening of the bamboo tubes using a rubber mallet. It is arranged chromatically like the marimba and has a range of one and a half octave. It is the bass of the ensemble and functions like a tuba.

Design-wise, the marimba used by BKME is similar to the western classical marimba. However, its keys are made of narra while the resonators are made of bamboo. It has a range of two octaves.

Gabbang is a bamboo xylophone, placed in a wooden box.

Pito is a transverse bamboo flute which came from the Igorot. It is the main melodic instrument of Musikawayan and has a range of two octaves. In the original Musikong Bumbong, there were four players playing this instrument. More *pito* players were added in the next generations of the band.

Bamban, a bamboo instrument that is similar to the kazoo, was invented by Mr. Ramos' great-grandfather. The actual instrument is made of *buho* (a species of bamboo), while the membrane is made from a piece shaved from the inner part of fresh kawayan. However, when the membrane gets wet, it can no longer be used and must be changed. From the original *bamban* developed by their ancestors, the Ramos family further developed the bamban and used cellophane to attach the membrane to produce a better sound and longevity.

Musikawayan also has a bamboo clarinet with a modern mouthpiece. Before, they used a reed and a mouthpiece made of bamboo, but when they tried to use a plastic mouthpiece from an alto saxophone, they found it easier to use.

Excerpt from the BMI Ethnography of Sta. Mesa's Banda Kawayan

"Concertized Bamboo Music Cultures"

By Patricia Marie U. Dizon

In terms of materials used in Philippine instrument-making cultures, bamboo is one of the commonly used. While the bumbong is often associated with the *musikong bumbong* bands in Bulacan, Calabig uses other types of bamboo. Apart from the *bumbong*, he often uses Chinese bamboo as he finds it more durable than the other varieties of bamboo. Known as *kawayang Tsina* in Paete, Laguna, this variety is also more malleable but less brittle when dry. *Buho*, a thinner variety of bamboo, is used too, albeit sparingly. Other varieties of bamboo that have thorns may be used in house construction, but not in instrument-making, as they are difficult to shape and become brittle when dried. Wood is used, with mahogany and acacia bearing the best sound.

The ensemble's instruments are the marimba, *angklung*, *panpipes*, *bumbong*, *kalatok*, *kalagong*, and *kiskis*.

Some of the instruments used by Banda Kawayan are quite familiar to the audiences. The marimba, for example, is one of the generally used musical instruments in the world, but while the standard marimba seemed bulky for ingress to and egress from the performance venue, Fred's version is more portable and foldable. "*Makikita ninyo yung k'wan, dapat collapsible pagkatapos dapat magaan pero yung durability hindi. Magagaan ang kahoy kong ginagamit, lalo na ngayon makikita mo yung nakasandal na 'yon*," he said. The portability of the bamboo marimba is something unique to the Banda Kawayan's set of musical instruments. Another distinct feature of Fred's version of the marimba is that the resonators are made of bamboo tubes, instead of the usual steel.

Apart from the marimba, some of Banda Kawayan's musical instruments which are commonly known are the *angklung* and the *panpipes*. The *angklung* is an Indonesian musical instrument which is played by shaking the lower part while supporting the upper part. Meanwhile, the *panpipes* are a bunch of small bamboo pieces put together and are played by blowing. In the Banda Kawayan, the *angklung* section plays the chords, while the *panpipe* section plays the ensemble's counterpoint. In terms of appearance and construction, while Banda Kawayan's version of the *angklung* looked almost the same as its Indonesian counterpart, it was a different case for the *panpipes*: Normally, *panpipes* are bound equally on top with varying sizes only towards the bottom part. Banda Kawayan's version, on the other hand, has a neat arrangement of the pipes at the bottom, but its front and back parts are bound unevenly. During the 1970s up to the 1980s, Fred made an attempt to have his *angklung* design patented. However, he did not renew it anymore, as registration for patented designs is not only costly but also tedious, as it had to be returned when the design did not meet the government's standards.

The rest of Banda Kawayan's musical instruments are rather distinct, which allowed both Fred and the ensemble to make their mark in the instrumental ensemble performance scene here in the Philippines and abroad.

First is the *bumbong*. Although similar to the commonly designed *panpipe* in shape (a set of hollow bamboo pieces) and manner of playing (by blowing), the *bumbong* has holes on the surface, as the instrument is meant to be blown on such portion. In

an ensemble, it is divided into two sections: the “bass” section, which plays the pieces’ lower wind parts and the “horn” section, which plays the higher wind parts.

Another distinct instrument of the ensemble is the *kalagong*. It has the same frame as the marimba, but the bamboo poles are hung inside the frame. Hence, the name “*kalagong*” is a portmanteau between “*kala*” (Calabig) and “gong,” which is the instrument maker’s brainchild. The *kalagong* is the ensemble’s bass parts.

Another instrument named after Fred is the *kalatok*, which is also a distinct musical instrument of Banda Kawayan. It is played similarly to the Latin American clave and the Kalinga pateteg by being struck by a stick. While the clave is a wooden block, and the pateteg a bamboo blade, the *kalatok* is a hollow bamboo tube with a resonator found on the surface and closed ends.

The *kiskis* is a scraper also made of Chinese bamboo. It is tuned through holes in between slits. Scrapers have been existent in other music cultures, especially Afro-Latin. Both the Afro-Latin percussive scrapers, known as the guiro, and Banda Kawayan’s *kiskis* are played by being scraped by another stick. However, while the guiro is gourd-shaped, the *kiskis* is rather elongated.

The *kalatina*, another instrument made by Fred, is a hung quill-shaped musical instrument with a sound hole at the center made of *buho*. It is played by being struck.

A pitched wooden block musical instrument made of Chinese bamboo, which still had no name, was used by Calabig during the group’s tour in Spain. It is played by striking such wooden block with a mallet.

Fred also makes wind musical instruments, such as flutes. While he sometimes creates Cordillera musical instruments, such as the *kubing* and *tongatong*, it is not as lucrative as the making of Banda Kawayan’s musical instruments previously stated. He also prefers to make solid-sounding musical instruments, which are able to resonate even with the absence of microphones onstage.

Some of the instruments are old, the bumbong being the oldest at twenty years. They are carefully handled, but in case of breakage, the instruments are wrapped with a tape, which do not affect the instruments’ sounds. Sometimes, they are repaired using Mighty Bond. However, when instruments are already difficult to repair, they are replaced.

The instruments are tuned at 440 Hz. Initially, they used a pitch pipe. However, due to technological advancements, they now use a chromatic tuner, which is widely available through Google Play and App Store downloads. One of the more modern tuner’s advantages is that a visual representation of such pitch is now shown on-screen, either on a tablet or on a smartphone.

Banda Kawayan’s musical instruments are stored in its headquarters inside the Communication Foundation for Asia (CFA) compound, which is owned by the Catholic Church. The group was able to acquire such place when they performed during the opening of the CFA compound in the 1970s. As the group performs during tours, Calabig always sees to it that the instruments are collapsible and fit inside the balikbayan box (30x20x10). He also ensures that the instruments would not exceed the maximum weight

required for hand carried luggage. The group's costumes are designed by renowned fashion designer, Renee Salud, who is the cousin of Fred's sister-in-law. Fred stated that Salud does not charge the group an exorbitant amount of money. "Kilalang-kilala ko si Mama Renee so, pakisamahan lang," he said.

Fred finds the musical instruments useful in classroom settings, especially the angklung and tongatong. Based on his experience, anyone who has both the reading and writing ability will be able to play such instruments. Having a concept of good musical timing is also essential.

Fred was able to pass on the bamboo musicmaking activity to his family, particularly to his firstborn child and second child. However, they were not able to adapt the instrument-making process, as it was too challenging, and it would require skills in carpentry and use of sharp tools (e.g., saw, *katam*). At present, his townmates, Jun Cadapan (a former Paete municipality employee) and Rodrigo Cadapan (a former trumpet player of the Philippine National Police Band), who are also skilled woodcrafters from Paete, Laguna, are the apprentices who help him in instrument-making. He is currently making a living by creating musical instruments for Far Eastern University and another one for Tarlac and Canada. Recently, the Nueva Vizcaya chapter of the United Methodist Church purchased a twenty-five-piece set of angklung from him, which costed Php 1,200 per piece. Such set contained chromatic pitches. According to Fred, he earns more money in instrument-making.

Excerpt from the BMI Ethnography of Dulag's Karatong Festival
"Dulag, Leyte: Presenting Identity through Bamboo Musical Instruments"
By Roan May D.G. Opiso

KARATONG FESTIVAL

Karatong Festival is the annual fiesta of Dulag, Leyte. It highlights the Waray bamboo percussion karatong, which in ancient times was used by the Dulagnons as an alarm when an impending danger or calamity would threaten their town (Cagara, interview, 19 October 2019).

Mr. Orlando Cagara, Secretary to the Sanggunian Bayan of Dulag, shared that Dulag's previous festivals used to be heavily patterned after the Sinulog of the neighboring province of Cebu. In 2019, however, the Dulag local government unit (LGU), under Honorable Mayor Mildred Joy Que, decided to impart their own identity in their festival. At the same time, they wanted an all-bamboo music featured in their festivity. Hence, they elected to develop the karatong and other bamboo musical instruments.

Why Bamboo?

Upon developing the new concept for their festival, the Dulag LGU looked into the resources of their land. Dulag, according to Cagara, is generally a flat land, and though they do not have a forest, the town has six to seven species of bamboo. They have *pattong*, *tangnan*, *kawayan*, *bagakay*, among others.

BAMBOO INSTRUMENTS

In preparation for the Karatong Festival, the Dulag LGU invited Prof. Jocelyn Guadalupe of the UP College of Music and Philippine Society of Music Educators and Mr. David Dino Guadalupe of the UP Center for Ethnomusicology to give a workshop on bamboo music. This workshop was attended by teachers of the different participating schools in the municipality and was later followed by rehearsals and further encouragement from the local government.

During the said workshop and the succeeding preparations for the festival, the schools were able to create different types of bamboo instruments. Additionally, the bamboo instrument makers named their creations based on their Leyte/Samar heritage, congruent with their aim of highlighting the Dulag identity in their festivities. Hence, the following instruments came to being.

Halo or Bayawak

The *halo or bayawak* is a set of three (3) long bamboo tubes laid in a row. Its name came from the Waray word halo, meaning bayawak in Filipino or monitor lizard in English, because its length is similar to that of the said animal.

The instrument is made from the *bolo* bamboo, which is locally known as *kayali*. The instrument is played by striking the holes at the end of the bamboo using a rubber mallet. (Some players improvised using a rubber slipper). Each bamboo tube of the *halo*

has its own note. The longer the length of tube, the lower the pitch. At the same time, the shorter the tube, the higher the pitch.

The *halo* has different sizes and each one is tuned to itself; meaning, a *halo* is not necessarily tuned the same way as other *halos*. Played in an ensemble, its function is comparable to that of the tenor drum of a marching band.

In creating the *halo*, Romeo and Paul of San Jose National High School first took the measurements of a bamboo pole. They then hollowed out the inside, and used a pitch pipe to get the pitch of each tube. Paul started with the lower 'do' before going up to the highest note. Mr. Go, another teacher from their school, made the decorative design of the instrument.

Karatong

As part of an ensemble, the karatong has a function similar to that of the snare drum of a marching band. Though it has been a traditional instrument in Dulag, the instrument makers in the workshop made innovations on its design to suit their needs. For example, they came up with a karatong version which has a handle on one end so that it can be held by one hand and beaten with a bamboo stick by the other hand. This style makes it convenient for the drummer to dance and to play the instrument at the same time. Another style of the *karatong* has the instrument placed on a stand, which is then strapped to the player. This set-up allows the instrumentalist to simultaneously dance and play the instruments using two sticks. A third version is a set of karatong grouped in three's and placed in a frame.

Like the functional innovations made on the instrument, the decorative design of the *karatong* also varies depending on the maker. In San Jose, for example, some of their karatong were styled like a fish.

Karatoktok

The Dulagnons who attended the bamboo music instrument seminar also made a bamboo lyre or xylophone which they called *karatoktok*. It is played using a bamboo stick as a mallet. Like the *halo*, the *karatoktok* is also made from *kayali*.

The *karatoktok* in San Jose NHS was tuned by Paul and used the notes of the song "Si Filemon" which they performed in the festival.

Other Instruments

Aside from the *halo*, *karatong*, and *karatoktok*, the instrument makers also made other different bamboo instruments. They have another bamboo xylophone similar to a *gabbang*, and a huge bamboo clapper. The bamboo clapper though was used more as a prop than an instrument.

They also have a smaller version of halo called by Romeo as *munting sawa*. These shorter bamboo tubes are grouped together (one version has at least 8 tubes) in one row and set vertically in a frame, unlike the *halo* which is laid down horizontally. The *munting sawa* was acquired after the Karatong Festival and is reserved to be played in future activities.

Instrument	Method of Sound Production	Melodic Range	Role in the Ensemble	Function in Ritual or Social/Cultural Activity
Halo (or bayawak)	Hole at the end of the bamboo tube is hit with a rubber mallet	Range depends on the maker (ex. length of the bamboo tube)	Functions like a tenor drum in a marching band; keeps the beat	Part of an ensemble, the instruments are played during fiestas and other celebrations, mostly functioning as entertainment music; accompaniment to street dancing.
Karatong	Body of bamboo is hit with a stick	-	Functions like a "snare drum"	Part of an ensemble, the instruments are played during fiestas and other celebrations, mostly functioning as entertainment music; accompaniment to street dancing. Traditionally used as a warning device.
Karatoktok	Bamboo slabs are hit with a bamboo stick	Instrument tuned to itself (range may vary depending on the song)	Plays melody	Part of an ensemble, the instruments are played during fiestas and other celebrations, mostly functioning as entertainment music; accompaniment to street dancing.
Bamboo xylophone (gabbang)	Bamboo slabs are hit with a stick	(No data)	(No data)	(No data)
<i>Munting sawa</i>	Hole at the end of the bamboo tube is hit with a rubber mallet	(No data)	(No data)	(No data)

Excerpt from the BMI Ethnography of Angono's Musikong Bumbong

"Musikong Bumbong: Musika Kawayan of Angono, Rizal"

By Juliet R. Bien

DESCRIPTION OF THE INSTRUMENTS

A. French horn

Simulating a brass French horn, the *musikong bumbong* prototype has the following measurements: length 20 inches, width 17 inches, height 3 inches, diameter of biggest barrel 2 inches, and valve 5 inches. It has three valve levers interconnected side by side by a 7-inch bamboo dowel. The "fake" mechanism rests on a lead pipe which coils around the body of the instrument spanning 51 centimeters. From the valves, the coiled pipe passes through a 3-inch bamboo connector with an approximate diameter of 1 inch. The pipe is then attached to the bigger vertical pipe (17 inches length and 3 inches diameter) which is the opening of the instrument. The air column spans the length of the mouthpiece traversing towards the main vertical chamber. The sound mechanism does not include the valves and the coiled pipes. The tone is produced through the transverse pipe towards the vertical chamber. The endpipe is slashed with 7-inch incisions to produce a buzzing sound. The endpipe functions as the bell valve. Unlike the brass French horn and the *musikong bumbong* instruments in other towns like Bulacan, the *bumbong* version of Angono's Musikang Kawayan does not have a bell.

B. Trombone

The trombone prototype is a composite of the main body and its removable slide. Like the other bumbong instruments, the bamboo design is a simulation of a brass trombone. The air is blown through the mouthpiece channeled towards the tip of the instrument, with the slide as decoration. The buzzer takes the place of the bell and the bell valve. The measurements are as follows: length (with unstretched slide) 42 inches, length of removable slide 24 inches, mounting rod for the removable slide 13 inches, total length of removable slide curved pipe 51 inches, removable slide brace 4 inches, length of main body 55 inches, incised buzzer 4 inches, main air chamber 19 inches, and airpipe diameter 1.75 inches.

The sound is produced by blowing the air through a detachable funnel-shaped mouthpiece carved from a piece of wood. The platform supports the main body of the instrument and holds the parallel rods from which the removable slide is mounted. The air column chamber traverses down to the left and curves toward the bell cylinder. The sound passes through a series of bamboo pipes with various lengths: a medium-sized vertical 7-inch chamber, a small U-shaped 8-inch pipe, 0.5 inch connected to a 2-inch medium sized joints, and a 28-inch barrel with a 2-inch diameter.

The removable slide consists of an elongated deep and narrow U-shaped bamboo pipe spanning 55 inches supported by two braces on both ends. The braces consist of a medium sized bamboo cylinder and a smaller pipe which are located closer to the mouthpiece and towards the end of the instrument, respectively. This mechanism is randomly slid and does not produce musical sounds. During performance, the slide is moved randomly as an imitation of how trombones are played in the brass band.

C. Trumpet

The trumpet is a singular piece of instrument with the following measurements: length 22 inches, width 7 inches, height approximately 5 inches, total length when uncoiled 56 inches, thickest diameter 1.25 inches. From the mouthpiece, the lead pipe loops into a curved body which consists of three interconnected bamboo pipes imitating the outline of a brass trumpet. The three parallel 7-inch valves strung together by a bamboo rod is mounted into the loop by two 5-inch braces on the left and right sides with a distance of two inches from the valves.

A narrow bamboo pipe is curved connecting the straight barrels towards the outlet. The largest pipe with a diameter of 1.25 inches takes the place of the bell. Like the other instruments of the set, the endpipe is a cylindrical piece of bamboo and incised towards the end to produce a buzzing sound. Air is simply blown into the mouthpiece and adjusted by the musician's lips according to the repertoire. The valves are randomly played to imitate trumpet-playing in the Western brass band tradition.

D. Euphonium

The largest instrument in the collection, the euphonium consists of two vertical pipes supported by two horizontal braces. The three valves constructed parallel to each other are connected by narrow pipes looping its way towards the lead pipe simulating the design of a brass euphonium. The measurements are as follows: longer vertical pipe right side 29 inches, horizontal beams 13 inches, shorter vertical pipe 14 inches, width 12 inches, length of coiled frame 50 inches, valves 7 inches, finger stop diameter 0.25 inches, and endpipe diameter 3 inches. The euphonium is the only instrument with a finger stop bored through the lower corner of the larger vertical pipe at the opposite side from its player.

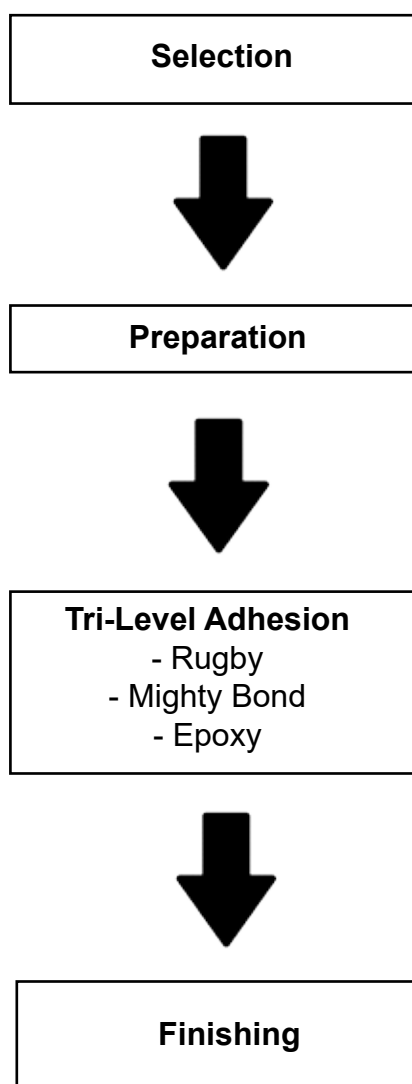
The simulation of its brass counterpart is done by constructing the coiled mechanism on top of the larger pipes functioning as the main frame supporting the valves and the coils. From the mouthpiece, the sound is produced passing through the horizontal beam, then through the curved pipe at its end as it connects to the endpipe. Like the rest of the instruments, the bell is replaced by the endpipe incised with 7-inch slits and produces a buzzing sound. The tones are refined by adjusting the finger stop. The rest of the coiled mechanism and the valves are not functional. Like the rest of the instruments, they are designed to imitate the brass euphonium but do not produce any sound.

The instrument's "aesthetic look" consists of the valves and the coils mounted on the bigger pipes. Between the first and the second valve, a narrow bamboo pipe is curved to form a U-shaped pipe towards the body of the shorter vertical pipe. A bigger U-frame enclosing the valves is constructed approximately two inches on the left side of the first valve whose end is attached to a bigger vertical pipe attached to the smaller vertical supporting pipe frame. A curved enclosure is completed by a U-shaped narrow pipe connected to the lead pipe.

Excerpt from the BMI Ethnography of Bulacan's Musikong Bumbong
"Mr. Rodrigo Anastacio and the Banda Santa Clara Musikong Bumbong
of Obando, Bulacan"
By Daniel Ray C. Eramis

PRODUCING THE BAMBOO MUSICAL INSTRUMENTS

Based on the discussion with Sir Rod, the diagram will explain the analyzed systematic process of producing the bamboo musical instruments from selection of materials to the final form of an instrument:



For Sir Rod, the best time to choose bamboo for an instrument is when it is fresh, so as it is easier to cut. He prefers to source his supply from Bulacan, usually from the areas of Plaridel, Pulilan, and Baliwag as it is available all year round within the province. He chooses *buho* (a hollow kind of bamboo) as small tube pieces for regular to small sized instruments and large bamboo species (size of saucer in diameter) for the bigger sized instruments like the *bajo* (tuba/bass).

After selecting, the specific parts of the bamboo are cut to fit the need of an instrument. They are then prepared by sanding out unnecessary parts and treating the insides with varnish to avoid weevil infestation.

Soon after the treatment had dried up, the bamboo parts are structured together and undergo three layers of adhesives:

- 1) Initially, the parts – inclusive of auxiliary parts – are joined with Rugby.
- 2) Once dried, the structure is reinforced with Mighty Bond. Sometimes small bamboo sheets or sticks are used to strengthen specific spots.
- 3) The joints are covered with Epoxy to fully strengthen the structure.

After the Epoxy has dried up, the excess is sanded off to provide aesthetic look for the finished instrument. Sometimes, if deemed necessary, certain areas are given another layer of adhesive. Once done, the whole instrument is protected with a layer of varnish.

For instrument making process and design patent concerns, the table below describes the innovative technique/s that Sir Rod had applied in his craft:

Instrument		General Bumbong Instruments	Folk Obando – Banda Santa Clara	Individual Sir Rod
Pitched	Flute (Recorder)	Made of bamboo	-	-
	Saxophone	Originally, all instruments were made of straight bamboo cuts, having varying sizes and lengths depending on the desired role.	Took shape with the use of cara-bao horn as reed, and tin can as resonator.	The most unique methods that Sir Rod utilized in producing the instruments are as follows: 1) Use of varnish to protect instruments from infestation. 2) Use of the trilevel adhesion method (rugby, mighty bond, and epoxy). 3) Use of rattan for the bells of specific instruments using them.
Buzz	Flute		Instruments began taking shape similar to their brass band counterparts with use of kazoo as primary producer of sound.	
	Transverse Flute			
	Trumpet			
	Clarinet			
	Baritone			
Trombone				
Single-Pitched	French Horn	Like all instruments in the ensemble, took shape into their brass band counterparts with the spit type sound production.		
	Bajo (Tuba)			

Aside from the innovations presented in the table above, some notable features of the bamboo musical instruments of Banda Santa Clara are the keys attached in the instruments like those in French Horns which do not really have any function but as accessories for performative purposes. They are made of button-shaped plastic attached to spring mechanisms to make them realistically look like their counterparts.

THE INSTRUMENTS AND THEIR ROLES

The ensemble of Banda Santa Clara Musikong Bumbong can be categorized under three methods of sound production: 1) Pitched Instruments, 2) Spit-type (like in brass instruments), and 3) Kazoo-based or Buzz Membrane. Due to the limitations of majority of the instruments, they play songs that are usually in or are transposed to the key of C Major. The table below helps to explain the methods of sound production, range, and role of each instrument in the ensemble.

Instrument		Method of Sound Production	Melodic Range	Role in the Ensemble
Pitched	Flute (Recorder)	- A kiss-flute method where a reed is fixed on the instrument - Pitch is manipulated by covering holes	c1 to c3 (2 octaves) but can be higher if advanced	Lead Voice
	Saxophone	- Using the carabao horn as reed and tin can as resonator - Pitch of the sound produced is manipulated by covering holes.	c1 to c3 (2 octaves) but can be higher if advanced	Lead Voice
Buzz Membrane	Trumpet	- Sound is produced by speaking through the mouthpiece which cause the membrane to buzz. - The pitch of the sound produced is based on the performers singing.	Range depends on the performer. Instrument can make clear sound from a to a2	Lead / 2nd Voice
	Flute & Transverse Flute			Lead / 2nd Voice
	Clarinet			2nd Voice
	Baritone			2nd / 3rd Voice
	Trombone			2nd / 3rd Voice
Single-Pitched	French Horn	Similar to brass instruments, The pressure of air affects resultant pitch	Between G to A (Overtone Possible)	Ostinato Response to Bass
	Bajo (Tuba)		Between C to D (Overtone Possible)	Down Beat Keepe
Percussion	Cymbals	Clashing (Metal)	-	For Emphasis Parts
	Snare Drum	Membrane and Rim Striking	-	Time Keeper, Signals the Band
	Bass Drum	Membrane Striking	-	

Excerpt from the BMI Ethnography of Maasin's Tultugan

"Tultug: The Root of Maasin's Tultugan Festival"

By Hermie F. Cartagena

INTRODUCTION AND HISTORICAL BACKGROUND

Bamboo plants are abundant in Maasin. One of the varieties that can be found in the area is the *"kawayan tinik"* and is produced yearly with more than 2 million poles. Thus, bamboo is the primary material in producing local products and handicrafts in Maasin. The place has been supplying bamboo poles and various handicrafts in Iloilo and has even been exporting them internationally (Marin, 2017). Being one of the primary exporters of bamboo products in the region and the country, this municipality in the central part of Iloilo became known as the bamboo capital of the Visayas.

In celebration of the abundant supply of bamboo products in the municipality, the Tultugan Festival was established. The festival, initiated or institutionalized by Mayor Mariano Malones in 1999, is considered a contemporary festival sponsored by the government (Muyco, 2016). Tultugan came from the word *"tultug,"* which means the act of making sound by tapping or striking the bamboo instrument (Marin, 2017).

The Tultugan Festival is primarily dedicated to bamboo and its various uses and functions. Held every fourth week of December, it features street dancing and a tribal dance competition. The highlight of the celebration is the tribal dance competition that showcases the various usage of bamboo in the community. The performers show how to creatively utilize the bamboo materials as part of the performance, incorporating them into their props, costumes, and instruments. Aside from the street dancing and tribal dance competition, the festival also highlights the *"Gwapo Karabaw,"* a contest wherein the participants adorn their water buffalos with various ornaments; *"Parada sang Litson,"* where roasted pigs are presented in the streets through a parade; *"Rara Amakan"* contest, wherein the speed in weaving bamboos is tested; *"Kadang Race,"* a test of balance; and the search for Miss Tultugan ("News Today" 2005, p. 1, paragraph 3).

The festival is a way of preserving the local culture for the next generations to come, and various approaches were applied in redesigning and modernizing the bamboo material. Likewise, various rhythmic patterns were composed and developed by local musicians to provide a dynamic and vibrant sound in the performance (Marin, 2017; Jacinto, 2013).

PRODUCING THE BAMBOO MUSICAL INSTRUMENTS

All bamboo instruments and products in the municipality of Maasin are made from the *kawayan tinik* variety. Ms. Rosemarie Bayag explained that the best time for the harvesting of the *Kawayan tinik* is from September to January. At this time, the bamboo is mature enough to be crafted as a musical instrument. The diagram below shows the systematic process of making a bamboo instrument from planning up to its final form.

Planning

What kind of instrument do you plan to make?



Bamboo Selection

- Found uphill
- Red pigmentation (mature bamboo)
- Tapping the bamboo pole or checking its resonance



Crafting of instrument

- Cutting of bamboo
- Making holes, removing nodes, and making slits in the bamboo



Refinement and finishing

- Sound quality/ resonance checking
- Durability checking

For Bayag, the first step in making a bamboo instrument is planning. She explained that an instrument maker should have in mind what kind of instrument he or she desires to make.

Secondly, the selection of bamboo must be fit to be crafted as an instrument. For Bayag, the best bamboo material are those bamboos that are situated uphill and have red pigmentations already. She also taps the bamboo pole using a bolo to check for sound quality. For her, not all bamboo has the potential to “sing.”

Bayag further explained the science behind this process. According to her, bamboos that are situated uphill have less moisture in their fibers when compared to those bamboos found in the lowland area. She also added that bamboos found uphill are used to dry or humid climates. With that, the bamboo fibers are denser and sturdier. Also, she advised to look for the red pigmentation of the bamboo, which signifies that it is already mature enough and ideal to be constructed as an instrument. She further explained that mature bamboos with denser fibers are better because they produce a brighter and better sound resonance or what she called “*nagakanta nga instrumento*” (singing instrument).

The third step is choosing the appropriate part of the bamboo fit to be an instrument. According to Bayag, she only cuts and uses the pusog and the puno part of the bamboo – found in the lower part of the bamboo stem. Such parts have thicker bamboo flesh compared to its other parts. This is followed by carefully and neatly cutting or removing some bamboo nodes (for pipes or pitched instruments) and making a slit on one side of the bamboo (for rhythmic types of instrument like *tultug*). Bayag makes sure that this step is meticulously done for it may affect the sound quality of the instrument.

Lastly, refinement of the instrument is done by checking the sound quality (resonance) and durability. Bayag explained that once the bamboo instrument got cracked or damaged during the refinement stage, it is simply replaced. For her, the sound quality of the bamboo will be affected if adhesive will be applied to fix the broken part of the bamboo.

THE INSTRUMENTS AND THEIR ROLES

The instruments in the municipality of Maasin can be classified into two – rhythmic and pitched. Rhythmic instruments produce sound by striking, stamping, and clashing bamboos. Pitched instruments, on the other hand, produce melodies by striking the surface or tubes of the bamboo. These instruments are set in three octaves tuned to the key of C. The table below explains the method of sound production, melodic range, role in the ensemble, and function in social/cultural activity.

Instrument	Method of Sound Production	Melodic Range	Role in the Ensemble	Function in Social/Cultural Activity
<i>Tultug</i>	<i>Patik</i> (striking) of the stick on the <i>tultug</i>		Pulse/ timekeeper	Although the instruments may vary in sizes, their function can be taken as one – an accompaniment to Tultugan Festival. As an ensemble, the instruments provide a festive atmosphere in the festival highlighting the flexibility of bamboo as an instrument.
<i>Bayong</i>	Stamping (rubbing coating at the bottom part of then bamboo)		Bass	
<i>Tala-tala</i>	A hand-held instrument, the sound produced by clashing the bamboo		Clapper/ cymbals	
<i>Paraspas</i>	Hand-held instrument; Clashing of the bamboo strips		Rattle/ clapper	
<i>Gabbang</i> (bamboo xylophone)	Striking the surface of the bamboo	3 octaves tuned in the key of C	Melody	
Bamboo Pipes	Graduated hallowed bamboos. Striking the bamboo tubes from the top covering the holes.	3 octaves tuned in the key of C	Additional texture; melody	
Mini Pipes	Graduated hallowed bamboos. From the top, a rubber is stamped to the bamboo covering the entire bamboo circumference. The sound comes out from the bottom. However, the instrument can be reversed for different sound quality. Rubber stamping is still done at the top of the bamboo instrument covering the entire hole.	3 octaves tuned in the key of C	Melody	

Appendix B – Photos

BAMBOO ORGAN OF LAS PIÑAS



BAMBOO RONDALLA OF DIPOLOG



Bandurria



Ramon Santos Bandurria



Octavina



Laud 1



Laud 2



Bass



Guitar



Marimba

BAMBOO UKULELE OF CEBU



HuniUkulele's "bamboolele" (soprano size, teardrop shape)



HuniUkulele's "bamboolele" (concert size, standard shape)



HuniUkulele's "bamboolele" (tenor size, round shape)

HIMIG KAWAYAN OF QUEZON CITY, LAGUNA, STA. MESA

Pangkat Kawayan of Quezon City



Alugtog (Angklung)



Bumbong



Tipanklung



Tunog Hagdan



Chord Rattle



Marimba



Sungkit

Banda Kawayan of Sta. Mesa Manila



Angklung



Bumbong



Bumbong



Panpipe



Kalagong



Marimba



Kalatok



Kiskis

Biñan Kawayan Music Ensemble / Musikawayan of Laguna



Bumbong



Angklung



Tipanklung



Marimba

KARATONG FESTIVAL OF DULAG, LEYTE



Karatong



Clapper



Karatoktok



Halo



Bayawak/Hagubhob

MUSIKONG BUMBONG OF ANGONO, RIZAL



Trumpet



Trombone



French Horn



Euphonium

MUSIKONG BUMBONG OF BULACAN

Clarinet



Trumpet



Transverse Flute



Trombone



Euphonium



Baritone



Bajo - Tuba



TULTUGAN MUSIC OF ILOILO



Gwapo Karabaw*



Kadang Race*



Tribal Dance Participant*



Tribal Dance Participant*



Tribal Dance Participant*



Musical Instruments of a Tribal Group*



Tultugan Instrumental Ensemble**



Street Parade Featuring the Paraspas**

*Photos of past Tultugan Festivals courtesy of Mr. Gilbert Marin ("My Beautiful Iloilo" blog, 2017)

**Photos from the 2019 Tultugan Festival, courtesy of Ms. Rosemarie Bayag

Appendix C – URL Links

Suggested YT URL links during the time of publication only

Description	Link
Module 1: Bamboo Organ of Las Piñas	
BMI Video/Audio of the Bamboo Organ demonstration by Armando Salarza	http://tiny.cc/BMI-Bamboo-Organ
Video about Leo Renier	http://tiny.cc/Leo-Renier
Module 2: Bamboo Rondalla of Dipolog	
Videos of Dipolog's Bamboo Rondalla from BMI documentation	http://tiny.cc/BMI-Dipolog-Rondalla
Videos of Jay Sarita's workshop from BMI documentation	http://tiny.cc/BMI-Sarita-Workshop
Module 3: Bamboo Ukulele of Cebu	
HUNI Soprano Ukulele	http://tiny.cc/YT-HuniUkuleles-Soprano
HUNI Tenor Ukulele	http://tiny.cc/YT-HuniUkuleles-Tenor
Module 4: Himig Kawayan of Quezon City, Laguna, Sta. Mesa	
Angklung – Himig Kawayan video tutorial	http://tiny.cc/YT-Angklung1
Indonesian Angklung	http://tiny.cc/YT-Angklung2
Angklung Performance	http://tiny.cc/YT-Angklung3
Marimba – Himig Kawayan video tutorial	http://tiny.cc/YT-Marimba1
Philippine Music: Dipolog Marimba Nais Ko	http://tiny.cc/YT-Marimba2
Banda Kawayan Plays Pinoy Music	http://tiny.cc/YT-Marimba3
Himig Kawayan, a video tutorial	http://tiny.cc/YT-Bumbong
Pangkat Kawayan	http://tiny.cc/YT-PangkatKawayan1 http://tiny.cc/YT-PangkatKawayan2 http://tiny.cc/YT-PangkatKawayan3
Musikawayan /D' Original Musikong Bumbong 1896	http://tiny.cc/YT-Musikawayan1 http://tiny.cc/YT-Musikawayan2 http://tiny.cc/YT-Musikawayan3 http://tiny.cc/YT-Musikawayan4
PUP Banda Kawayan/Banda Kawayan Pilipinas	http://tiny.cc/YT-BandaKawayan1 http://tiny.cc/YT-BandaKawayan2 http://tiny.cc/YT-BandaKawayan3
Bocaue Central School Musikawayan	http://tiny.cc/YT-Bocaue1
Musikawayan ng DepEd, Bocaue	http://tiny.cc/YT-Bocaue2
Piano Chords Tutorial	http://tiny.cc/YT-PianoChords
Basic Articulation	http://tiny.cc/YT-BasicArticulation

Module 5: Karatong Festival of Dulag, Leyte	
BMI documentation of Labiranan Festival of the 2019 Karatong Festival by San Jose National High School, Dulag Leyte	http://tiny.cc/BMI-Dulag-Karatong
BMI documentation of "Commitment to Peace" speech choir presentation by the Youth of Dulag	http://tiny.cc/BMI-Dulag-Choir
Module 6: Musikong Bumbong of Angono, Rizal	
BMI interview with Prof. Jose Rommel "Jopat" Gragera, conductor of Angono Wind Ensemble, and Angono Bamboo Instrument Maker	http://tiny.cc/BMI-Gragera-Interview
Musiko 2013 #14 Angono Wind Ensemble	http://tiny.cc/YT-AngonoWindEnsemble
Magtanim Ay 'Di Biro Pinoy BK Channel	http://tiny.cc/YT-MagtanimAyDiBiro1
Magtanim Ay 'Di Biro by The Dawn	http://tiny.cc/YT-MagtanimAyDiBiro2
How to Play the Rainstick	http://tiny.cc/YT-HowToPlay-Rainstick
DIY Rainstick (Fun Kid's Craft!)	http://tiny.cc/YT-DIY-Rainstick
Italian making a rhythmic instrument out of bamboo	http://tiny.cc/YT-Rhythmic-Instrument
Module 7: Musikong Bumbong of Bulacan	
BMI interview with Rodrigo Anastacio, Bulacan Bamboo Instrument Maker	http://tiny.cc/BMI-Bulacan-Anastacio
BMI documentation of Marcelo H. Del Pilar students playing bumbong at Hiyas ng Bulacan 2020, Malolos, Bulacan	http://tiny.cc/BMI-Bulacan-MHDP
How do brass instruments make a sound	http://tiny.cc/Brass-Instruments
The Rig - How to Make a Hosepipe Horn	http://tiny.cc/YT-Hosepipe-Horn1
How to make a hosepipe horn	http://tiny.cc/YT-Hosepipe-Horn2
Hosepipe and orchestra	http://tiny.cc/YT-Hosepipe-Orchestra
Garden Hose Concerto	http://tiny.cc/YT-Hose-Concerto
Gaano Ko Ikaw Kamahal sung by Celeste Legaspi	http://tiny.cc/YT-GaanoKolkawKamahal
Module 8: Tultugan Music of Iloilo	
YouTube videos on Tultugan Festival	http://tiny.cc/YT-Tultugan-Festival1 http://tiny.cc/YT-Tultugan-Festival2 http://tiny.cc/YT-Tultugan-Festival3 http://tiny.cc/YT-Tultugan-Festival4

Appendix D – Music Scores

Module	Title
Bamboo Organ of Las Piñas	“Suite Cortesana No. 2”
Bamboo Rondalla of Dipolog	“Air on a G String”
Himig Kawayan of Quezon City, Laguna, Sta. Mesa	“Rikiting-kiting” “Pambansang Awit”
Musikong Bumbong of Angono Rizal	“Magtanim Ay ‘Di Biro”
Tultugan Music of Iloilo	“Cundansoy”
(Additional pieces)	“Bahay Kubo” “Balay sa Buntod”

Suite Cortesana No. 2

I. Sinfonia

Anonymous

The musical score is written for a single instrument in 3/4 time. It consists of five systems, each with a treble and bass staff joined by a brace. The notation includes various musical symbols such as notes, rests, accidentals, and repeat signs. The first system has five measures. The second system has five measures, with a repeat sign at the end. The third system has five measures. The fourth system has five measures, with a repeat sign at the end. The fifth system has five measures, with a repeat sign at the end.

Transcribed by David Dino S. Guadalupe

Air on a G string

for Rondalla

Johann Sebastian Bach

♩ = 78

Bandurria

Octavina/Laud

Laud/Guitar

Guitar/Bass

14

17

transcribed by David Dino S. Guadalupe

Rikiting-kiting

Isinaayos ni: Genaro Cesar Edilo Jr.

Isinaayos: 27 Nobyembre 2015

[A]

The musical score for "Rikiting-kiting" is presented in three systems, each containing four staves for different instruments: Gabbang Marimba, Angklung, Guitar, and Bumbong. The music is in 3/4 time.

System 1 (Measures 1-7): The key signature changes to one sharp (F#) and the time signature changes to 3/4. The first staff (Gabbang Marimba) has a key signature change to one sharp and a common time signature change to 3/4. The second staff (Angklung) has a key signature change to one sharp and a common time signature change to 3/4. The third staff (Guitar) has a key signature change to one sharp and a common time signature change to 3/4. The fourth staff (Bumbong) has a key signature change to one sharp and a common time signature change to 3/4. The first measure of the first staff is marked with a key signature change to one sharp and a common time signature change to 3/4. The first measure of the second staff is marked with a key signature change to one sharp and a common time signature change to 3/4. The first measure of the third staff is marked with a key signature change to one sharp and a common time signature change to 3/4. The first measure of the fourth staff is marked with a key signature change to one sharp and a common time signature change to 3/4. The first measure of the first staff is marked with a key signature change to one sharp and a common time signature change to 3/4. The first measure of the second staff is marked with a key signature change to one sharp and a common time signature change to 3/4. The first measure of the third staff is marked with a key signature change to one sharp and a common time signature change to 3/4. The first measure of the fourth staff is marked with a key signature change to one sharp and a common time signature change to 3/4.

System 2 (Measures 8-14): The first staff (Gab. Mrb.) has a key signature change to one sharp and a common time signature change to 3/4. The second staff (Ang.) has a key signature change to one sharp and a common time signature change to 3/4. The third staff (Gtr.) has a key signature change to one sharp and a common time signature change to 3/4. The fourth staff (Bum.) has a key signature change to one sharp and a common time signature change to 3/4. The first measure of the first staff is marked with a key signature change to one sharp and a common time signature change to 3/4. The first measure of the second staff is marked with a key signature change to one sharp and a common time signature change to 3/4. The first measure of the third staff is marked with a key signature change to one sharp and a common time signature change to 3/4. The first measure of the fourth staff is marked with a key signature change to one sharp and a common time signature change to 3/4.

System 3 (Measures 15-21): The first staff (Gab. Mrb.) has a key signature change to one sharp and a common time signature change to 3/4. The second staff (Ang.) has a key signature change to one sharp and a common time signature change to 3/4. The third staff (Gtr.) has a key signature change to one sharp and a common time signature change to 3/4. The fourth staff (Bum.) has a key signature change to one sharp and a common time signature change to 3/4. The first measure of the first staff is marked with a key signature change to one sharp and a common time signature change to 3/4. The first measure of the second staff is marked with a key signature change to one sharp and a common time signature change to 3/4. The first measure of the third staff is marked with a key signature change to one sharp and a common time signature change to 3/4. The first measure of the fourth staff is marked with a key signature change to one sharp and a common time signature change to 3/4.

22 [B]

Gab. Mrb.

Ang.

Gtr. 22 C G G G7 C

Bum. 22

29

Gab. Mrb.

Ang.

Gtr. 29 G7 C G G G7

Bum. 29

36

Gab. Mrb.

Ang.

Gtr. 36 C

Bum. 36

Pambansang Awit ng Pilipinas

Isinaayos ni:
Genaro Cesar Edilo Jr.
22 Pebrero 2014

The musical score is written for four instruments: Gabbang Marimba, Angklung, Gitara, and Bumbong. The key signature is one sharp (F#) and the time signature is 2/4. The score is divided into three systems, each containing staves for the four instruments. Measure numbers 10, 19, and 28 are indicated at the start of their respective systems. The first system (measures 1-9) includes a first ending bracket over measure 9. The second system (measures 10-18) includes a second ending bracket over measure 18. The third system (measures 19-27) includes a third ending bracket over measure 27. The notation includes various musical symbols such as treble and bass clefs, key signatures, time signatures, and specific instrument labels. The score features a mix of eighth, quarter, and half notes, as well as triplets and chords. The Bumbong part is written in the bass clef, while the other instruments are in the treble clef. The Angklung part uses a simplified notation with vertical lines and dots to represent the instrument's sound. The Gitara part includes chord symbols (G, D, C) and specific fretting instructions. The Gabbang Marimba part uses a simplified notation with vertical lines and dots to represent the instrument's sound. The Angklung part uses a simplified notation with vertical lines and dots to represent the instrument's sound. The Bumbong part uses a simplified notation with vertical lines and dots to represent the instrument's sound.

Magtanim ay di Biro

Philippine Folk

Mag - ta - nim ay di bi - ro mag - ha - pong na - ka - yu - ko Di man

6
lang ma - ka - u - po. Di man lang ma - ka - ta - yo. Bra - so

10
ko'y na - ma - man - hid. Bay - wang ko'y na - nga - nga - wit. Bin - ti

14
ko'y na - mi - min - tig sa pag - ka - ba - bad sa tu - big. Sa u -

18
ma - gang pag - gi - sing ang la - hat i - i - si - pin. kung sa -

22
an may pa - ta - nim may ma - sa - rap na pag - ka - in. Bra - so

26
ko'y na - ma - man - hid. Bay - wang ko'y na - nga - nga - wit. Bin - ti

30
ko'y na - mi - min - tig sa pag - ka - ba - bad sa tu - big. Ha -

34
li - na, ha - li - na ma - nga ka - li - yag. Ta - yo'y mag - si - pag - u - nat - u - nat.

42
Mag - pa - ni - ba - go ta - yo ng la - kas pa - ra sa a - raw ng bu - kas.

Transcribed by David Dino S. Guadalupe

Cundansoy

Con - dan - soy, i - nom tu - ba La - loy. Di -

4
li ko mo - i - nom, tu - ba pa - it as - lom

8
Ang tu - ba sa bay - bay, pa - ten - te mo a - ngay

12
ta - lac - san mo diu - tay po - no ang ma - la - way

Transcribed by David Dino S. Guadalupe

Bahay Kubo

Traditional
arr.for FPRDI
Bamboo Ensemble
by D&J Guadalupe

$\text{♩} = 54$

Bamboo Marimba

Bamboo Tube

Bamboo Sticks

F G C C G7 F C C

11

BB Mrm.

BB Tube

BB stks

G G C C G7 F C F G C

21

BB Mrm.

BB Tube

BB stks

C G7 F C C G G C C

31

BB Mrm.

BB Tube

BB stks

G7 F C F G G

39

BB Mrm.

BB Tube

BB stks

C C G7 F C C

45

BB Mrm.

BB Tube

BB stks

G G C C G7 G7

51

BB Mrm.

BB Tube

BB stks

C F G G C C

Balay sa Buntod

(Visayan children's song)

May ba-lay ga-may sa bun - tod Kang ta-tay na-nay kad - tong ti-nu-kod

5 Dm G7 C Am Dm G7 C F
May ba-tang ga-may na - min - ta - na bo - u - tan si - ya nga ba - ta

9 C Am F C F G C Am
Ang i-yang na-nay gar - bu - sa Nag - la-kat - la-kat sa kal - sa-da

13 Dm G7 C Am Dm G7 C Repeat
Nag - si-naw si-naw ang pi - ta - ka pag an - do wa - la'y kwar - ta

transcribed by David Dino S. Guadalupe

**DOST Bamboo Musical Instruments Innovation Research and Development Program,
Project 1 - Documentation of Philippine Bamboo Musical Instruments,
UP Center for Ethnomusicology**

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